Midterm Exam Practice Questions (Weeks 1-7)

155-8056 XML and Web Services

Instructions: Each question has five choices (A-E). Only one choice is correct.

Week 1

1. What does the acronym XML stand for?

A) Executable Markup Link
B) Extended Module Language
C) Extensible Markup Language
D) External Mapping Layer
E) Encoded Machine Logic

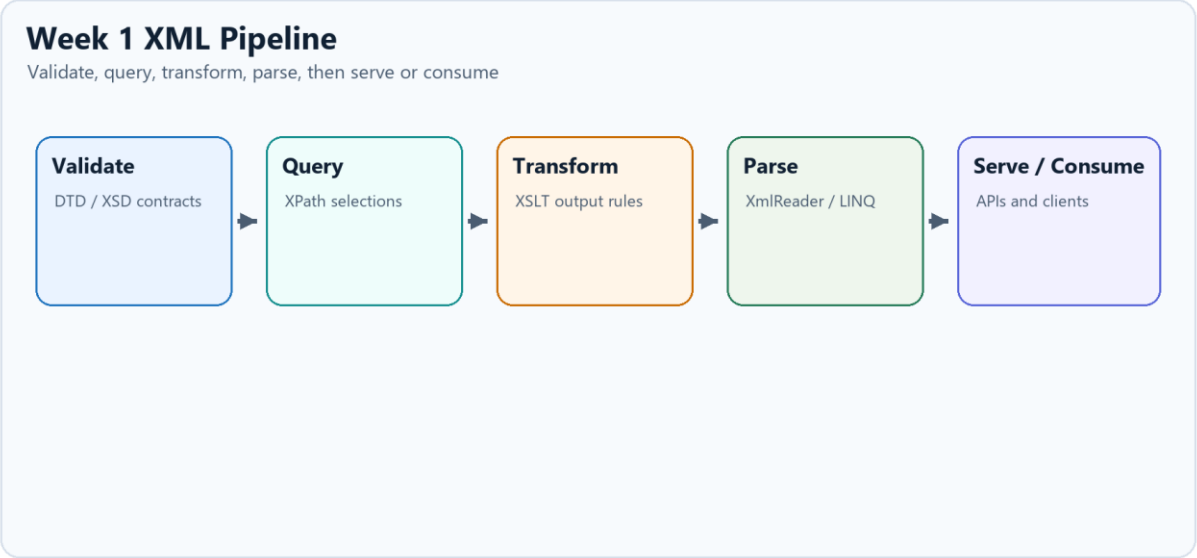
2. According to the lecture, which reason best explains why XML is still important today?

A) XML is still common in enterprise integration
B) It always uses less bandwidth than JSON in real deployments
C) It replaced every modern data format used on the web
D) It can be processed only inside web browsers and nowhere else
E) It removes the need for parsers, validators, and tooling

3. Which item was listed as a required tool or skill for this course workflow?

A) A Linux-only web server with production deployment access
B) An Oracle database license with enterprise support enabled
C) A dedicated SOAP appliance for packet-level XML routing
D) A mobile emulator for Android and iOS interface testing
E) .NET SDK support through Visual Studio or another compatible setup

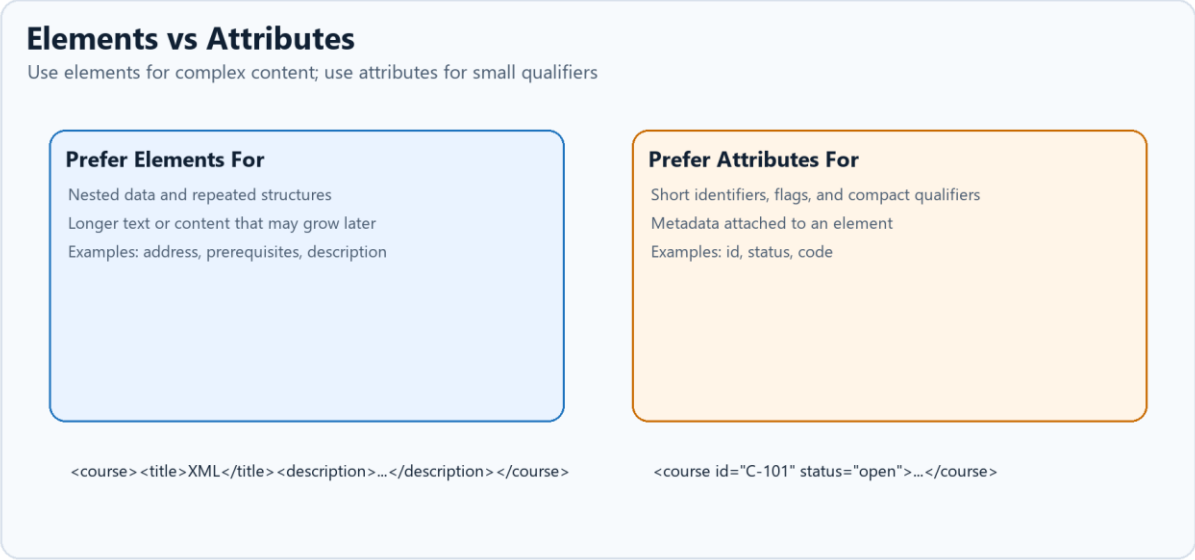
4. In the course pipeline diagram, which stage comes immediately after Validate?



A) Transform

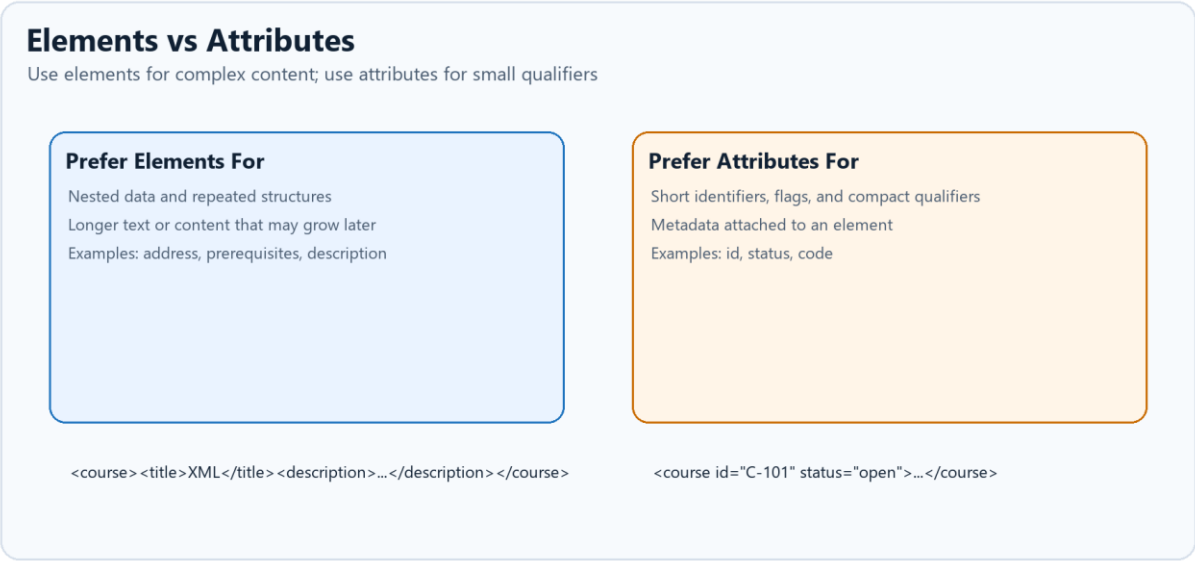
B) Query
C) Serve/Consume
D) Parse
E) Serialize

5. When should elements usually be preferred over attributes in XML design?



A) When the value must always stay hidden from readers
B) When the data must stay a single short character
C) When the value should never repeat across the document
D) When the data may later become nested or repeatable
E) When namespaces are forbidden by project policy

6. In the elements-versus-attributes diagram, which use case fits attributes better than elements?



A) A long product description with paragraphs
B) A repeating list of course prerequisites
C) A short flag or identifier on an element
D) A nested address structure

E) Mixed text with markup inside it

7. Which rule is mandatory for every well-formed XML document?

A) It must contain exactly one root element
B) It must always reference an external schema
C) It must avoid attributes completely
D) It must include a namespace prefix on every element
E) It must be served over HTTP

8. Why is the following XML not well-formed?

```
<note><text>Tom & Jerry</text></note>
```

A) The root element name is illegal
B) The text node is too short
C) The ampersand forces JSON mode
D) The text must be wrapped in CDATA in all cases
E) The ampersand is unescaped inside normal text

9. What is the problem with using <Root></root> in the same XML document?

A) XML ignores closing tags
B) XML is case-sensitive, so the names do not match
C) A capital letter is illegal in XML names
D) The slash must appear in the opening tag instead
E) The document becomes JSON automatically

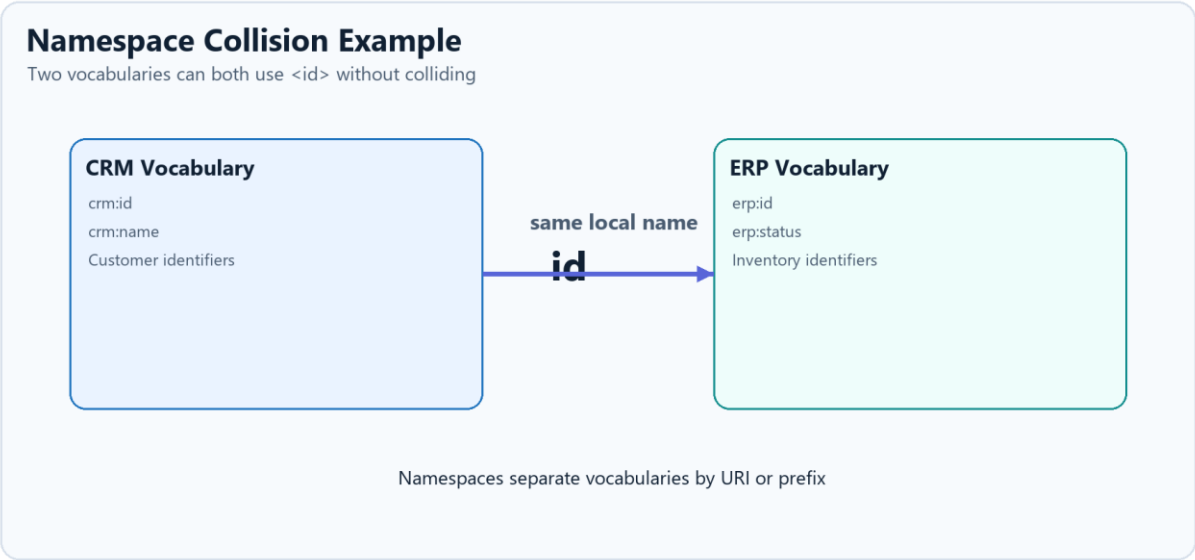
10. What is the main purpose of a CDATA section?

A) To declare a namespace prefix for later XML elements
B) To compress the XML document before transmission
C) To replace the root element during validation
D) To include text that should not be parsed as markup
E) To force the document to become valid against a schema

11. Which practice is safest when inserting user text that may contain special characters into XML in C#?

A) Concatenate raw strings and trust the user input
B) Remove every ampersand manually and ignore the rest
C) Let the XML library escape user text for you
D) Convert the text to uppercase first
E) Disable all closing tags

12. In the namespace collision diagram, what is the main role of XML namespaces?

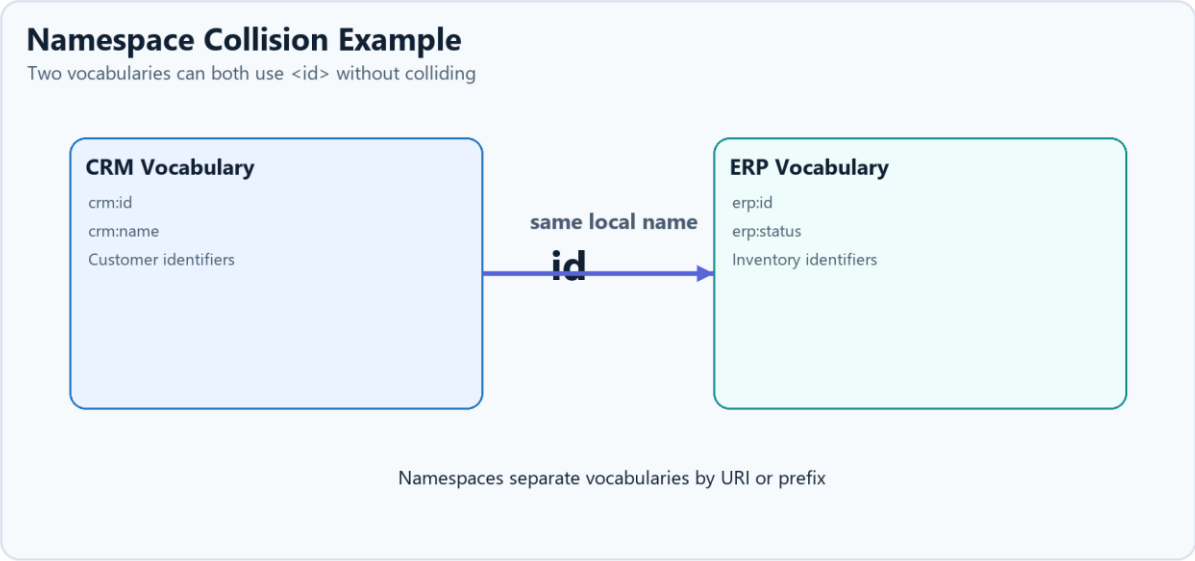


A) They prevent vocabulary name collisions
B) They make every document shorter
C) They remove the need for parsers
D) They are used only for comments
E) They guarantee schema validity by themselves

13. Why do queries often fail when a default namespace is present but ignored in code?

A) Default namespaces delete all attributes before parsing
B) Default namespaces work only in JSON-oriented clients
C) Element order changes silently whenever a prefix is missing
D) The XML becomes invalid automatically as soon as a namespace exists
E) Unprefixed element names still use the default namespace

14. Two XML vocabularies both use an <id> element. What problem do namespaces solve here?



A) They remove closing tags from one XML vocabulary
B) They separate identical names by vocabulary
C) They force every value to become numeric across documents

D) They convert matching elements into compact attributes
E) They make HTTP optional during XML processing

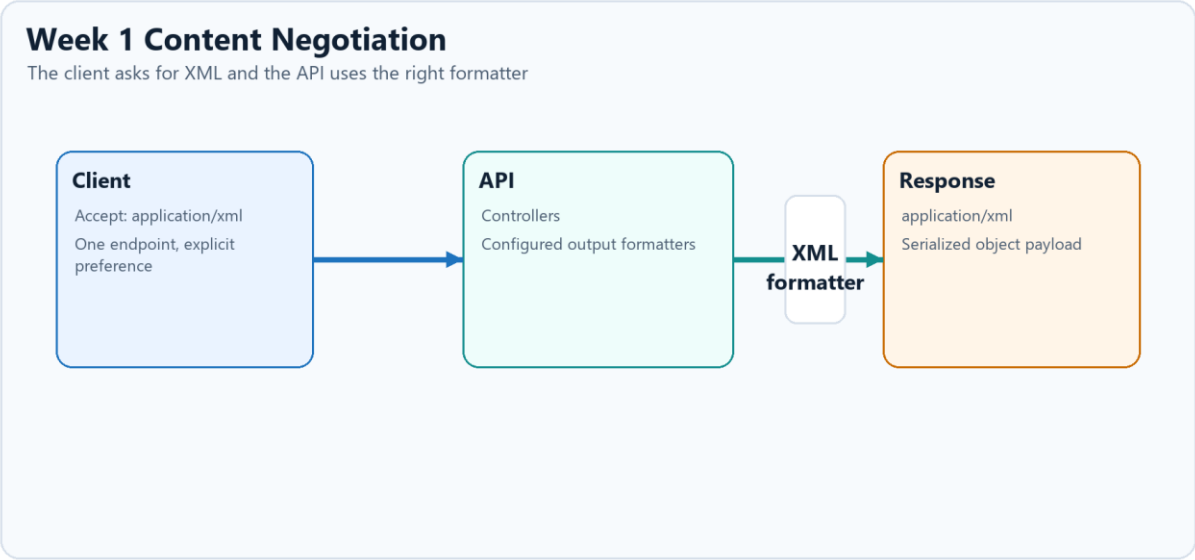
15. Which scenario fits XML better than JSON according to the pragmatic comparison in the lecture?

A) A flat object with only two simple scalar fields
B) A value that must never be nested or repeated
C) A payload with no need for structure, hierarchy, or metadata
D) A document-style payload that may need mixed content or markup
E) A case where text parsers and serialization libraries are forbidden

16. Which statement from the lecture is true about common document formats such as .docx and .xlsx?

A) They are plain JSON folders
B) They cannot contain structured markup
C) They are zipped XML-based packages
D) They are valid only on Linux
E) They cannot be generated by software

17. In the content negotiation diagram, what does the HTTP Accept header communicate in the Week 1 API discussion?

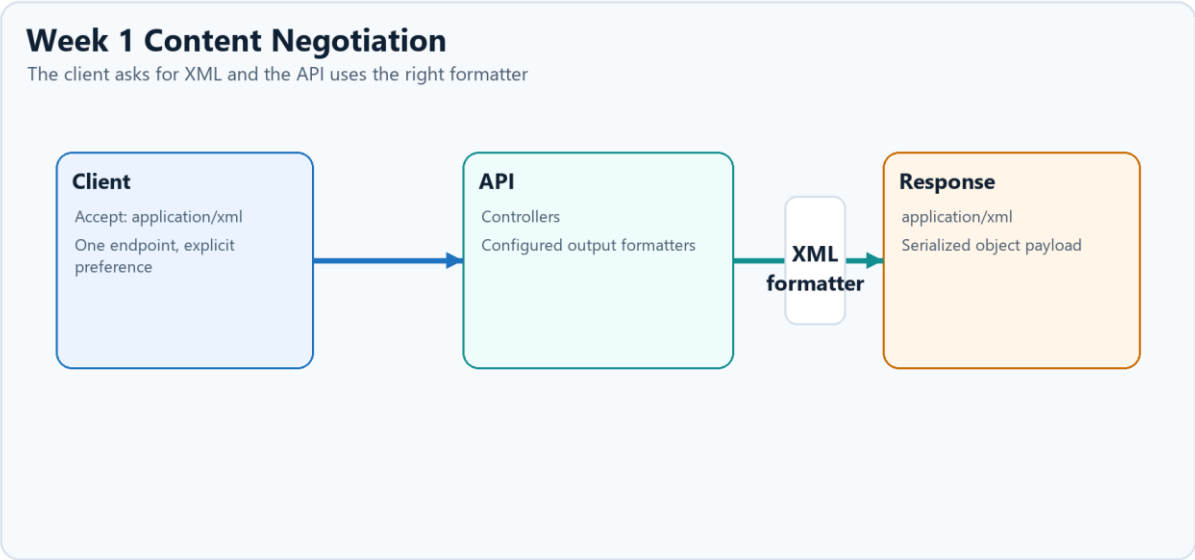


A) Which formats the client is willing to accept
B) Which server port should be opened
C) Which XML tags are required in the payload
D) Which controller class will compile
E) Which C# namespace is imported

18. What must be added conceptually so an ASP.NET Core endpoint can return XML from the same controller action?

A) A second web server written in PHP
B) A DTD declaration inside Program.cs
C) A new TCP protocol
D) A browser extension
E) XML formatters in the API pipeline

19. According to the content negotiation flow diagram, what does the API rely on to turn the response into application/xml?



A) A SQL trigger
B) An XML formatter
C) A ZIP extractor
D) A browser cookie
E) A DNS record

20. Why would a student set the Accept header in Postman, curl, or a C# client during Week 1 testing?

A) To rename the controller at runtime before the request is handled
B) To attach a DTD automatically to every HTTP response body
C) To encrypt the XML body without changing server code
D) To request XML from the same endpoint
E) To disable serialization and return raw memory objects

21. Which statement about self-closing tags is correct?

A) They are forbidden in XML
B) They can appear only inside CDATA
C) They are valid when an element has no content
D) They always create two sibling elements
E) They remove the need for a root node

22. Which character must be escaped if you want to use it literally inside normal XML text?

A) <
B) _
C) /
D) :
E) "

23. Which XML snippet demonstrates overlapping tags?

A) <a></a>
B) <a></a>
C) <a attr="1"></a>
D) <a><![CDATA[test]]></a>
E) <a></a>

24. Why does the lecture warn against stuffing real content into attributes?

A) Attributes are not supported by XML parsers
B) Attributes do not fit nested or repeating content
C) Attributes always force binary encoding
D) Attributes cannot contain letters
E) Attributes automatically create namespaces

25. Which project type matches the in-class "Hello XML API" demonstration?

A) A static HTML-only site with no backend
B) A Windows Forms desktop app
C) A mobile app with no HTTP support
D) An ASP.NET Core Web API using controllers
E) A database migration project

26. Which statement best captures the idea of content negotiation from Week 1?

A) A client must create a separate endpoint for every media type it wants
B) XML responses never come from APIs that use modern frameworks
C) One endpoint can return different formats based on headers and formatters
D) Serialization is unrelated to HTTP headers and formatter selection
E) Only browsers can request XML from a web API endpoint

27. Which statement is true about well-formed XML?

A) It means the XML syntax is structurally correct
B) It guarantees that business rules are satisfied
C) It requires an XSD file in every case
D) It allows multiple root elements if namespaces differ
E) It is a JSON compatibility mode

28. What does a default namespace apply to unless otherwise stated?

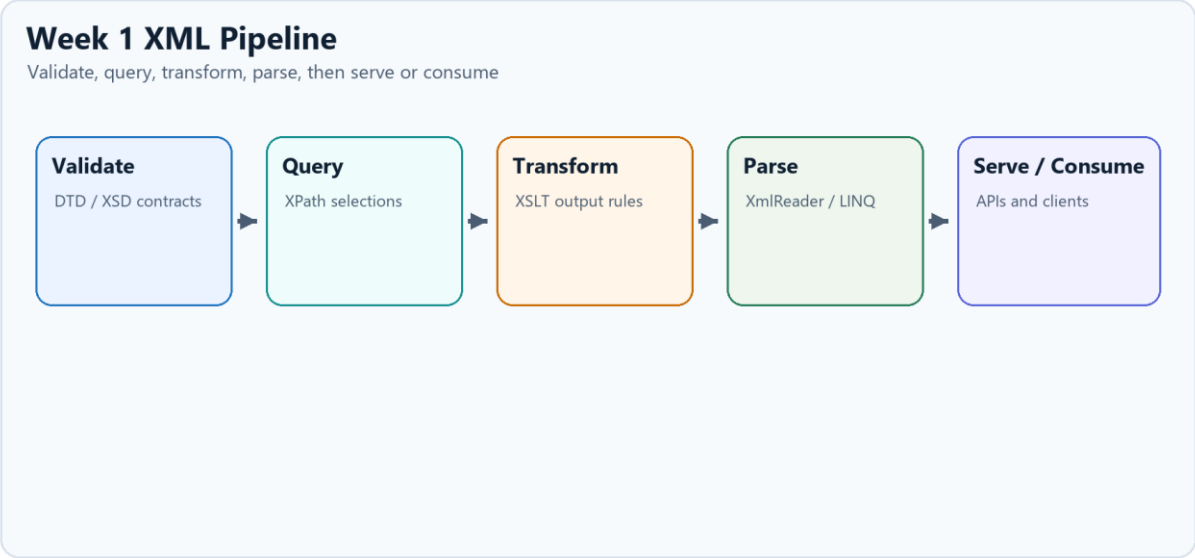
A) Only comments
B) Only processing instructions
C) Only numeric values
D) Only prefixed attributes
E) Unprefixed element names

29. Which core XML rule is violated in the following snippet?

```
<Root><Child/></root>
```

A) Attributes must be quoted
B) Start and end tag names must match exactly
C) CDATA must wrap every child element
D) Namespaces must be declared on every line
E) The root must be empty

30. In the Week 1 XML pipeline diagram, which stage is associated with XSLT?



A) Validate
B) Query
C) Serve/Consume
D) Transform
E) Compile

31. Which of the following is itself an XML-based vector graphics format?

A) PNG
B) GIF
C) SVG
D) BMP
E) MP3

32. Which Week 1 lab design choice is most aligned with the lecture guidance?

A) Use one clear root and consistent naming in a small domain XML
B) Start with multiple unrelated root elements from the beginning
C) Place all real content in attributes instead of structured elements
D) Avoid namespaces under every condition, even when names collide
E) Write malformed XML first and keep it as the final version

33. Which header value best expresses that a client wants XML back from an API?

A) Host: application/xml
B) Content-Type: only/xml
C) Authorization: xml
D) Cache-Control: xml
E) Accept: application/xml

34. Which snippet is well-formed XML?

A) <a></a>
B) <root><item id="7" /></root>
C) <root><item></root>
D) <one/><two/>
E) <root attr=7></root>

35. Which combination best summarizes the main Week 1 focus stated in the lecture?

A) XSD restriction facets, substitution groups, and validator tooling
B) XPath functions, sorting, and grouping logic
C) XSLT templates, parameters, and text output
D) Well-formed XML, namespaces, and a small API that can return XML
E) XmlReader subtree skipping and safe streaming settings

Week 2

36. Which statement correctly describes the relationship between well-formed and valid XML?

A) Every well-formed XML document is automatically valid
B) Every valid XML document must also be well-formed
C) Validity replaces the need for syntax rules
D) Well-formedness matters only when no root element exists
E) They are unrelated concepts

37. What is the main job of a DTD in an XML workflow?

A) To convert XML into JSON before validation begins
B) To encrypt element values during document loading
C) To replace the parser with a manual text editing process
D) To define the document's allowed structure and attributes
E) To remove the root element when content gets too large

38. In the internal-versus-external DTD diagram, where is an internal DTD subset written?

A) Inside the XML document's DOCTYPE
B) Only inside a separate .json file
C) Inside an HTTP response header
D) Inside a C# enum definition
E) Inside a namespace URI

39. When is an external DTD subset especially useful?

A) When XML is forbidden from using reusable external files
B) When a document should never be reused by any system
C) When many XML documents reuse one shared .dtd contract
D) When attributes must be disabled for every element node
E) When JSON schemas are being validated instead of XML rules

40. In the DTD content models diagram, what does the declaration <!ELEMENT book (title,author,year)> require?

DTD Content Models

Sequence, choice, and repetition operators

Pattern	Meaning	Example
(a,b,c)	Exact order	title, author, year
(a|b)	One choice	email or phone
a*	Zero or more	book*
a+	One or more	book+
a?	Zero or one	magazine?

A) Only one of the three child elements may appear under book
B) The child elements may appear in any order under book
C) The book element must be empty except for attributes
D) Attributes must replace the child elements completely
E) The children must appear in the sequence title, author, year

41. In a DTD content model, what does (email|phone) express?

A) The element is empty and may not contain child nodes
B) A choice where one of the listed alternatives is allowed
C) A sequence in one fixed order with both values present
D) A numeric range with lower and upper bounds
E) A namespace declaration for prefixed element names

42. What does the plus sign mean in a declaration such as book+?

A) Exactly zero books
B) Exactly one book
C) At most one book
D) One or more book elements
E) A book element with text only

43. What does the question-mark operator mean in a DTD content model?

A) The element may appear zero or one time
B) The element must appear at least twice
C) The element must contain only attributes
D) The parser should ignore the element
E) The element becomes a namespace prefix

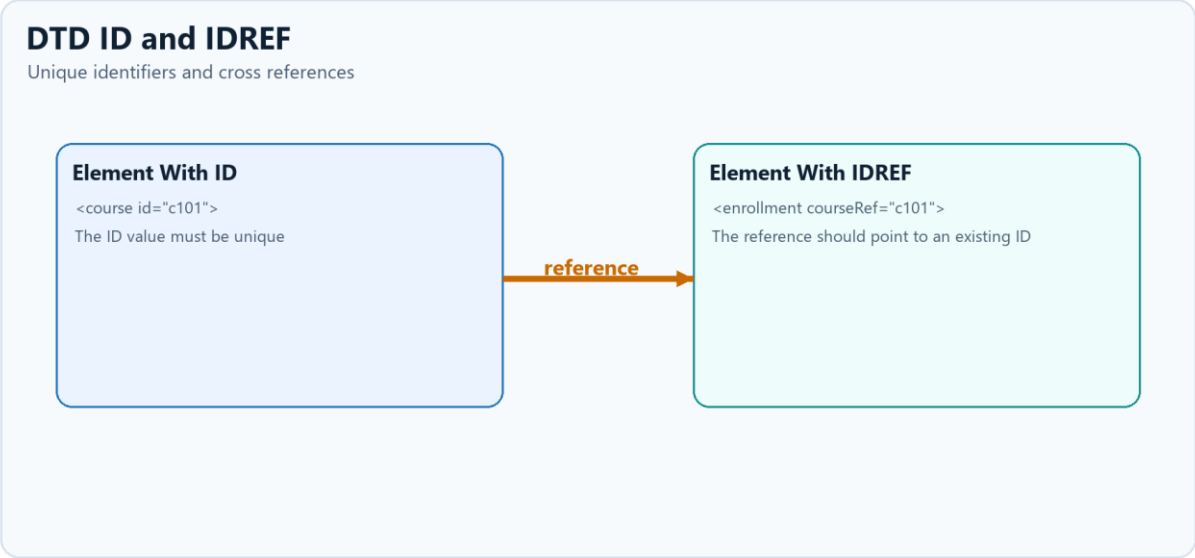
44. What does (#PCDATA) represent in a DTD declaration?

A) A required namespace prefix for element qualification
B) A binary attachment stored directly in the DTD
C) Parsed character data used as text content
D) A fixed attribute value applied during validation
E) A comment block that the parser preserves as data

45. What is the purpose of an ID attribute type in DTD?

A) It stores raw XML markup without escaping
B) It allows duplicate identifiers freely
C) It forces the value to be decimal only
D) It disables validation for the element
E) It defines a unique identifier within the document

46. In the DTD ID-and-IDREF diagram, what is an IDREF attribute used for?



A) To declare an HTTP endpoint for the XML resource
B) To reference an existing ID elsewhere in the document
C) To store CDATA sections directly inside attribute values
D) To mark an element as empty during validation
E) To replace namespaces with plain attribute names

47. What does #REQUIRED mean in an <!ATTLIST> declaration?

A) The attribute may appear at most once
B) The attribute is generated from JSON
C) The attribute is always numeric
D) The attribute must be supplied on that element
E) The attribute may never appear

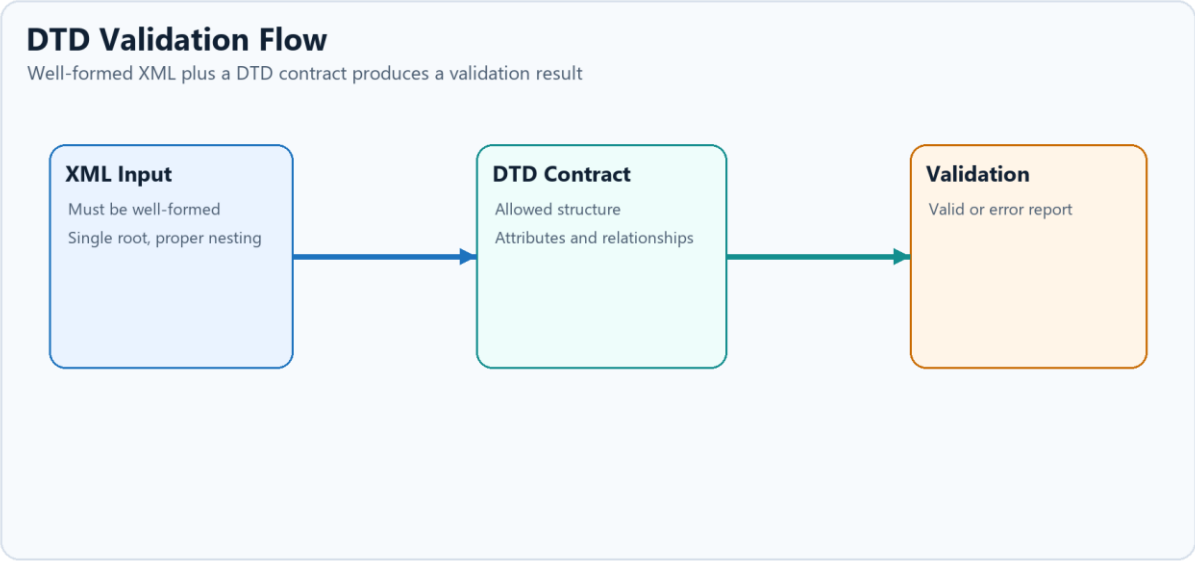
48. What does #IMPLIED mean in an <!ATTLIST> declaration?

A) The attribute is optional
B) The attribute is always inherited from the root
C) The attribute value must be an integer
D) The attribute must use CDATA only
E) The attribute creates a new element

49. What does #FIXED "student" mean for an attribute declaration?

A) The attribute must be removed from the document
B) The attribute becomes optional and random
C) The attribute is constrained to the fixed value student
D) The attribute can contain any value starting with student
E) The attribute is converted into an element

50. According to the validation flow diagram, which box supplies rules and structure before the validation result is produced?



A) API Gateway
B) XPath Query
C) CDATA Parser
D) JSON Contract
E) DTD Contract

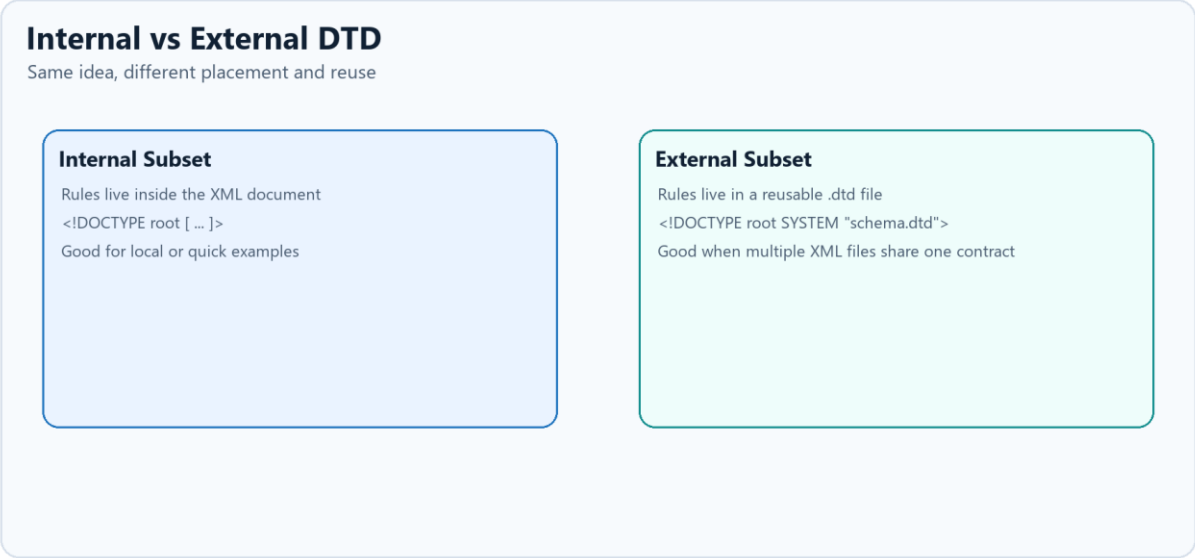
51. Why can a well-formed XML payload still break a downstream system?

A) Well-formed XML cannot contain attributes or text nodes
B) It may still violate the contract's structure or attribute rules
C) Well-formed XML is always encrypted before it is sent
D) Downstream systems read only JSON and reject every XML file
E) Namespaces stop working after parsing reaches the root element

52. A student includes all required child elements but puts them in the wrong order. Under a sequence declaration, what is the result?

A) The document becomes empty
B) The parser silently reorders the elements
C) The document becomes well-formed but never parsable
D) It may be well-formed but DTD-invalid
E) The DTD is ignored automatically

53. Which statement best compares internal and external DTD subsets?



A) External subsets are better for reusable shared rules
B) Internal subsets can only define attributes, not elements
C) External subsets make XML stop being well-formed
D) Internal subsets always validate faster than external ones
E) Only external subsets support IDREF

54. Which DTD limitation was emphasized as a reason for moving toward XSD in Week 3?

A) DTD can work only on Linux-based XML servers
B) DTD cannot describe any structure, order, or attributes at all
C) DTD is weakly typed compared with XSD
D) DTD forbids text nodes completely in every valid document
E) DTD has no syntax for elements, attributes, or content models

55. Which additional weakness of DTD was highlighted in the lecture?

A) It cannot reference external files in reusable validation workflows
B) It always requires a web browser to validate XML documents
C) It is stronger than XSD for regex and datatype constraints
D) It stores XML as binary data instead of normal text markup
E) Its namespace handling is less precise than XSD

56. Where are DTDs still realistically encountered?

A) Only inside video codecs used by streaming platforms
B) In legacy integration systems and long-lived document pipelines
C) Only in mobile game engines for graphics metadata
D) Only when using relational databases and SQL views
E) Only in spreadsheet formulas created by end users

57. When validating XML against a DTD in .NET, which validation mode is the relevant one to configure?

A) ValidationType.JSON
B) ValidationType.SchemaOnly
C) ValidationType.XPath
D) ValidationType.DTD
E) ValidationType.Binary

58. What information makes a DTD validation error report more useful in practice?

A) Line and position details together with a readable message
B) Only the file size in kilobytes and the file name
C) The XML rendered as an image inside the editor
D) The browser history of the developer who opened the file
E) The color of the editor theme used during debugging

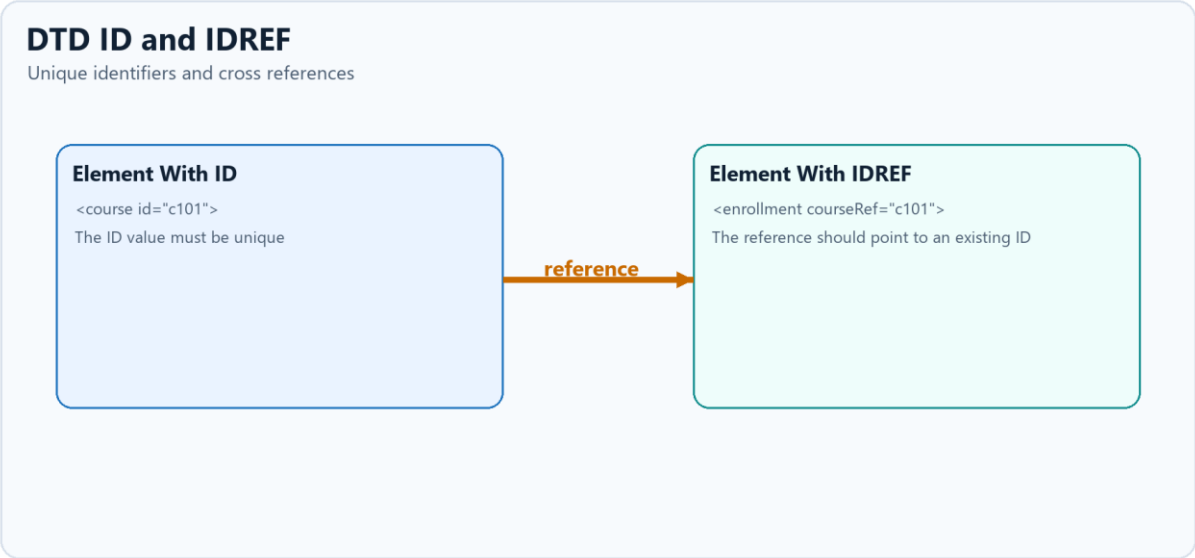
59. Why does the assignment-ready checklist ask for both valid and invalid XML instances?

A) Because invalid XML is easier to grade than valid XML
B) Because validators can never process correct examples
C) Because it proves the validation rules were actually tested
D) Because DTD works only on broken documents
E) Because namespaces are optional only in invalid files

60. Which DTD declaration keyword is used to define attributes for an element?

A) <!CHOICE>
B) <!GROUP>
C) <!CONTENT>
D) <!OPTION>
E) <!ATTLIST>

61. Which pair of DTD attribute types is designed for unique identifiers and references?



A) CDATA and ENUM
B) ID and IDREF
C) TEXT and VALUE
D) MIN and MAX
E) ATTR and NODE

62. Which statement is correct if an XML document has the right elements but they appear in the wrong sequence for the DTD?

A) The document becomes valid automatically because all names exist
B) The document cannot even be parsed as XML text
C) The wrong sequence has no effect on validation
D) The document may still be well-formed while failing validation
E) The parser changes the sequence to match the DTD

63. Which DOCTYPE form corresponds to an internal DTD subset?

A) <!DOCTYPE root [...]>
B) <!SCHEMA root { ... }>
C) <root dtd="inline">
D) <!DOCTYPE [root]>
E) <!DTD root internal>

64. Which example shows a reusable external DTD reference?

A) <root schema="rules.dtd">
B) <!DOCTYPE root EXTERNAL rules.dtd>
C) <!DOCTYPE root SYSTEM "schema.dtd">
D) <!ATTLIST root dtd CDATA "schema.dtd">
E) <root dtdref="schema.dtd">

65. Why is DTD often described as text-centric or weakly typed?

A) It stores all XML as images before validation occurs
B) It supports stronger date and decimal constraints than XSD does
C) It ignores element names entirely during content checking
D) It validates only JSON payloads instead of XML documents
E) It lacks the rich primitive types and detailed facets available in XSD

66. Which symbol means zero or more occurrences in a DTD content model?

A) +
B) *
C) ?
D) |
E) ,

67. Which symbol means one or more occurrences in a DTD content model?

A) ?
B) *
C) ,
D) +
E) |

68. Which debugging habit best matches the Week 2 lecture?

A) Compare the XML carefully against the DTD rules
B) Delete the DTD so the validation errors disappear immediately
C) Convert everything to JSON before checking any XML rules
D) Assume validation errors come only from namespaces and nothing else
E) Ignore line and position details while debugging parser output

69. Which DTD content model enforces the exact order title, then author, then year?

DTD Content Models

Sequence, choice, and repetition operators

Pattern	Meaning	Example
(a,b,c)	Exact order	title, author, year
(a|b)	One choice	email or phone
a*	Zero or more	book*
a+	One or more	book+
a?	Zero or one	magazine?

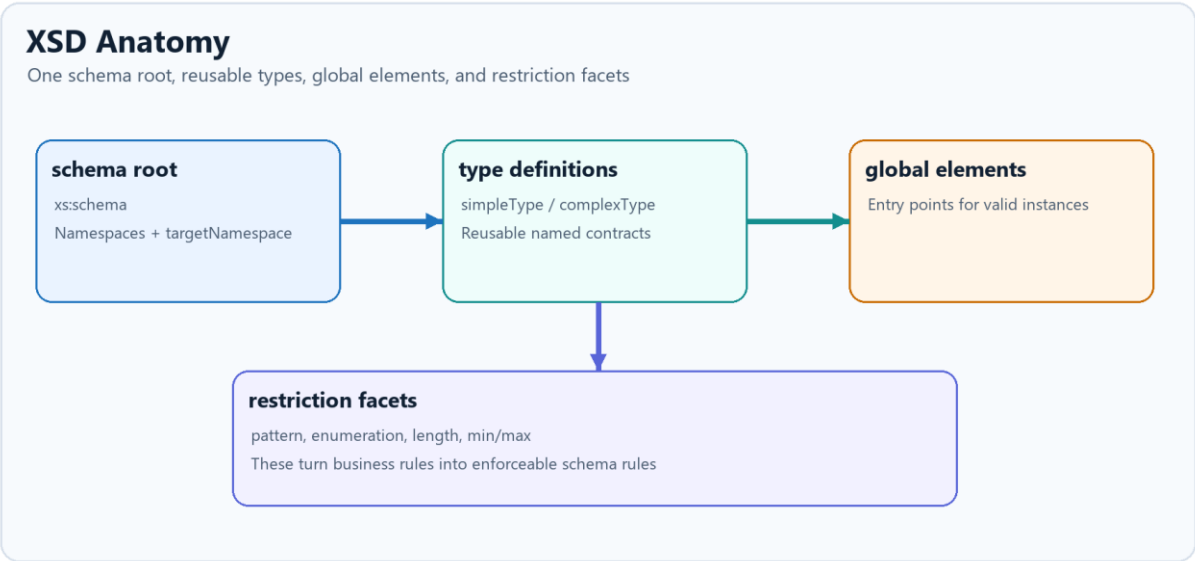
A) (title|author|year)
B) (title*)(author*)(year*)
C) (title,author,year)
D) (title?author?year?)
E) (year,author,title)

70. Which statement about valid XML is correct?

A) It may break syntax rules if the business meaning is correct
B) It never needs a declared contract
C) It is identical to plain text
D) It ignores required attributes when the root exists
E) It must be well-formed and follow the DTD rules

Week 3

71. In the XSD structure diagram, which labeled area contains pattern, enumeration, min/max, and length constraints?

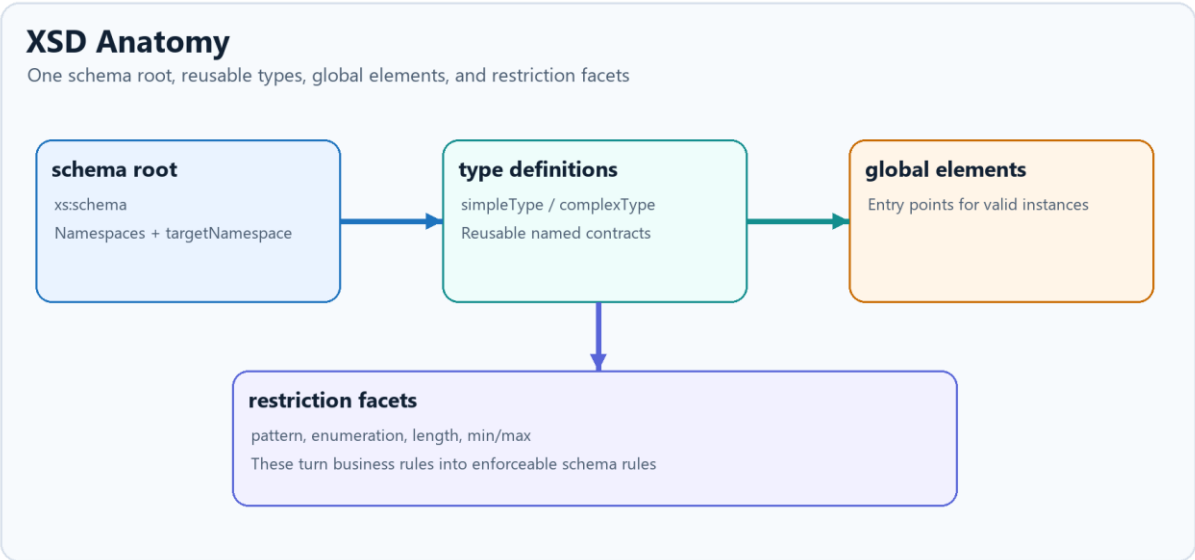


A) restriction facets
B) global elements
C) namespace aliases
D) document serializer
E) output formatter

72. What does an XSD primarily do for XML instances?

A) It turns XML into binary packets
B) It replaces element names with indexes
C) It defines legal structure and value rules
D) It forces all content into attributes
E) It removes the need for parsing

73. In the XSD anatomy diagram, what is the role of xs:schema in an XSD file?



A) It is only a comment wrapper around existing XML text
B) It is the schema root for namespaces and top-level rules
C) It is a special XPath function used in instance queries
D) It is the runtime XML instance itself rather than the schema
E) It is used only for CDATA and mixed content output

74. Why does targetNamespace matter in XSD design?

A) It makes all values lowercase before validation begins
B) It converts attributes into child elements automatically
C) It allows comments to be validated as business data
D) It identifies the vocabulary and prevents name collisions
E) It removes the need for global elements and type definitions

75. What is the practical effect of setting elementFormDefault="qualified"?

A) Attributes must become integers during instance validation
B) Only the root element keeps a namespace after parsing
C) Local elements are ignored during validation checks
D) The schema stops using targetNamespace for child elements
E) Local instance elements must be namespace-qualified

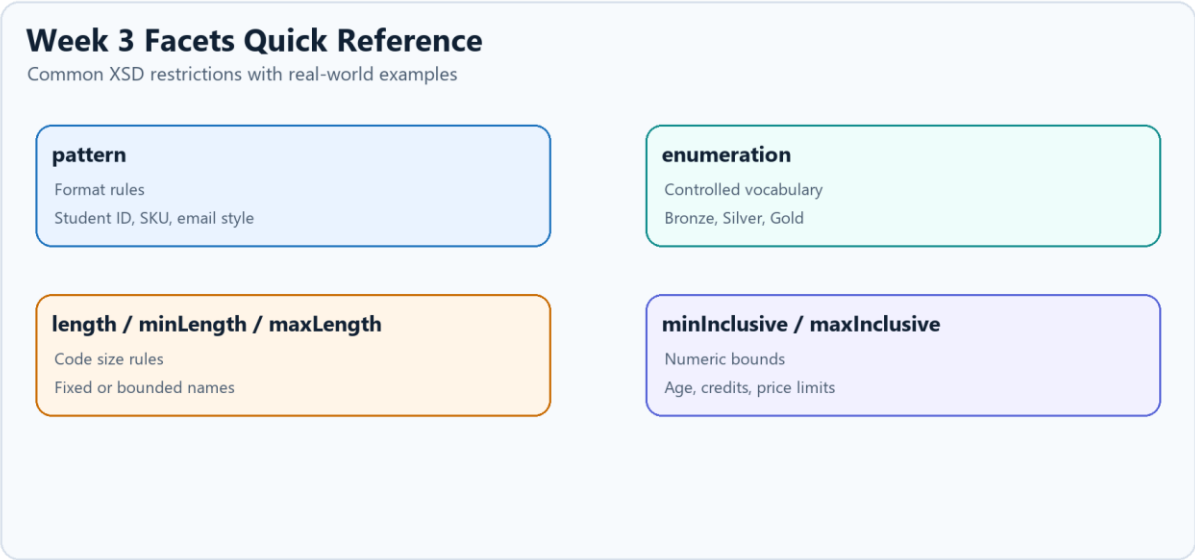
76. Which XSD primitive type best fits a whole-number count such as course credits?

A) xs:integer
B) xs:boolean
C) xs:dateTime
D) xs:anyURI
E) xs:base64Binary

77. Which XSD type is designed for true/false values?

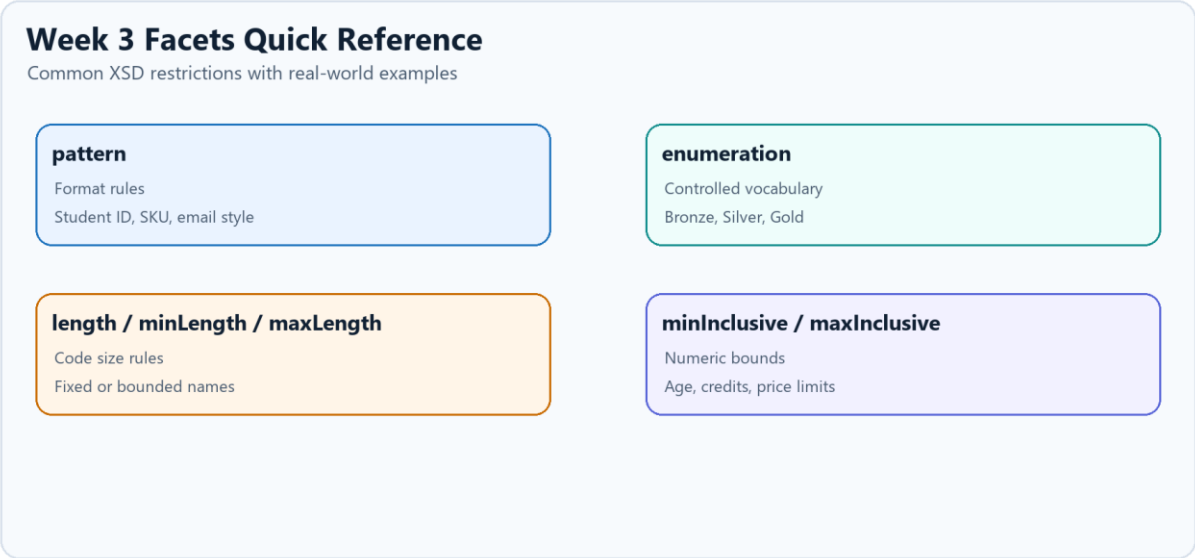
A) xs:token
B) xs:float
C) xs:boolean
D) xs:duration
E) xs:QName

78. Which XSD facet is used to limit a value to a predefined set such as Bronze, Silver, or Gold?



A) pattern
B) enumeration
C) minInclusive
D) fractionDigits
E) maxLength

79. In the Week 3 facets quick-reference diagram, which facet is most appropriate for enforcing a format like STU-2026-001?



A) totalDigits
B) enumeration
C) whiteSpace
D) pattern
E) maxExclusive

80. Which facet sets a legal minimum numeric value in XSD?

A) length
B) minLength
C) enumeration
D) pattern
E) minInclusive

81. If a code must contain exactly 8 characters, which facet is the most direct choice?

A) length
B) maxLength
C) minInclusive
D) enumeration
E) choice

82. Why are named reusable types useful in XSD?

A) They stop XML from using namespaces
B) They force every element to become optional
C) They reduce duplication and keep rules consistent
D) They replace global elements with comments
E) They remove the need for validation events

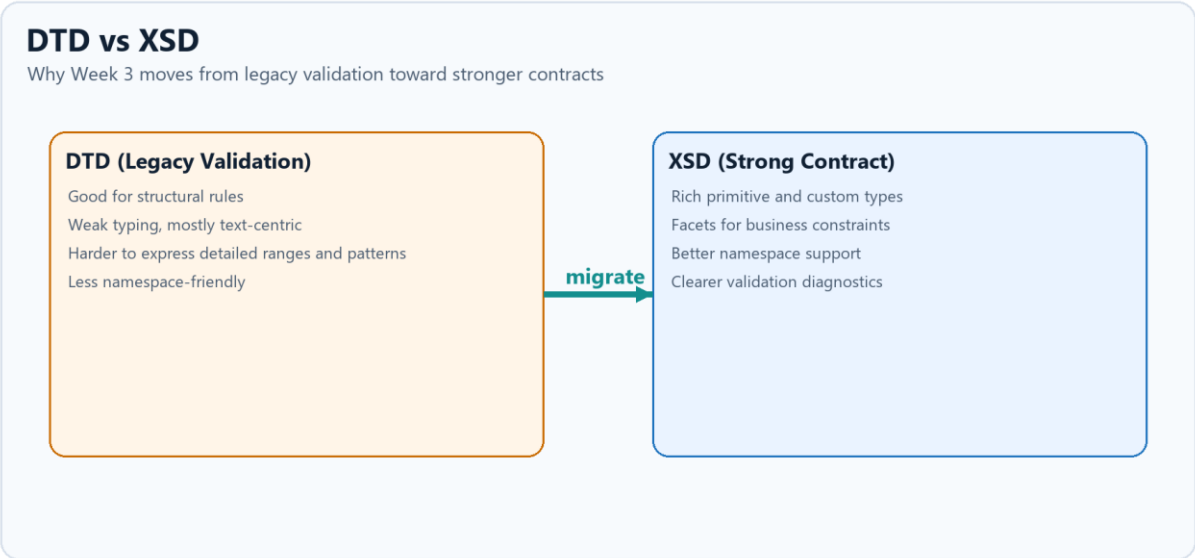
83. Which problem does targetNamespace help solve?

A) It shortens all XML values automatically during parsing
B) It prevents name collisions across shared vocabularies
C) It disables child elements in schemas with reusable types
D) It converts XML into JSON arrays for web APIs
E) It makes schemas unnecessary once a root element exists

84. If the schema requires qualified local elements, what happens when the instance omits the namespace on child elements?

A) The parser silently adds the namespace to every child node
B) The values are converted to attributes before validation
C) The instance becomes JSON-compatible and skips schema checks
D) Validation fails because the children are in the wrong namespace
E) The schema switches to DTD mode and ignores namespaces

85. In the DTD-versus-XSD comparison diagram, what major advantage does XSD provide over DTD?



A) It has no syntax for structure or repeated content
B) It removes the need for namespaces in shared vocabularies
C) It validates only comments and processing instructions
D) It supports fewer data types than classic DTD validation
E) It offers richer typing and stronger constraints

86. What do restriction facets accomplish in practical schema design?

A) They enforce business rules directly in the schema
B) They disable namespace checking for local elements
C) They remove the root element from validation logic
D) They replace XML instances with XSD files during processing
E) They convert every field to CDATA before parsing

87. Which XSD primitive type is best for a monetary amount such as tuition or price?

A) xs:boolean
B) xs:date
C) xs:decimal
D) xs:QName
E) xs:hexBinary

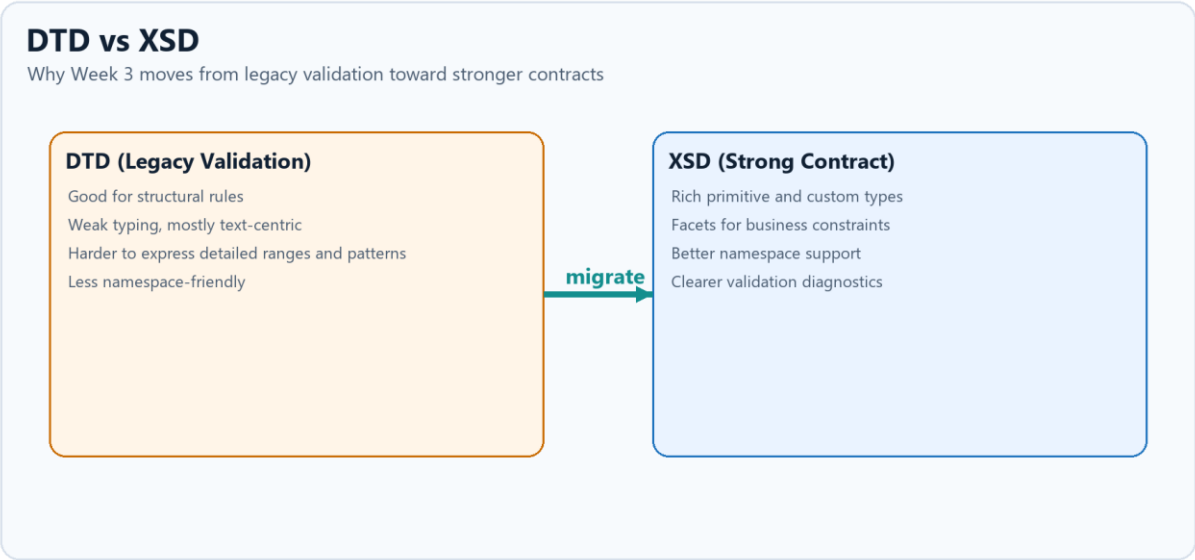
88. According to the lecture's modeling guidance, when should elements usually be preferred over attributes?

A) When the data must stay hidden from validators and tools
B) When the content may later become nested or repeated
C) When namespaces should be removed from the schema design
D) When XML should avoid text nodes and child structures
E) When only fixed values are allowed in every case

89. Which scenario fits an attribute better than a child element?

A) A repeating list of student achievements
B) A full mailing address with several nested parts
C) A long narrative description
D) A small qualifier such as status code or ID
E) A mixed-content paragraph

90. In the DTD versus XSD comparison diagram, which side is associated with clearer validation diagnostics?



A) Neither side provides diagnostics
B) The DTD side only
C) Both sides equally, with no difference
D) The JSON side
E) The XSD side

91. Why does the lecture recommend one clear namespace per vocabulary?

A) It improves interoperability and validation
B) It makes XML independent from schemas
C) It forces child order to become random
D) It disables all local element declarations
E) It converts prefixes into attributes

92. Which facet is appropriate when a field must be one of a small allowed vocabulary?

A) pattern
B) fractionDigits
C) enumeration
D) minLength
E) all

93. Which pair of facets is suitable for keeping an age between 18 and 65 inclusive?

A) length and maxLength
B) minInclusive and maxInclusive
C) enumeration and pattern
D) choice and sequence
E) whiteSpace and list

94. Why is enforcing constraints in XSD often better than checking every value manually inside application code?

A) Because schemas remove the need for programs and services
B) Because XML parsers refuse manual checks inside applications
C) Because attributes cannot be validated in code paths at all
D) Because the contract stays reusable and close to the data
E) Because XSD runs only after the database has accepted the row

95. What was the core goal of the Week 3 DTD-to-XSD conversion theme?

A) To remove all validation from the course workflow
B) To replace XML with HTML reports for every exercise
C) To shorten every element name used in the dataset
D) To avoid using namespaces forever in schema design
E) To move to stronger schema typing and better diagnostics

96. What do global elements act as in an XSD?

A) Entry points for valid XML instance documents
B) Comments that are ignored during validation
C) Replacement names for namespaces
D) Binary blobs embedded in the schema
E) Only temporary debug helpers

97. Which statement correctly distinguishes simpleType from complexType?

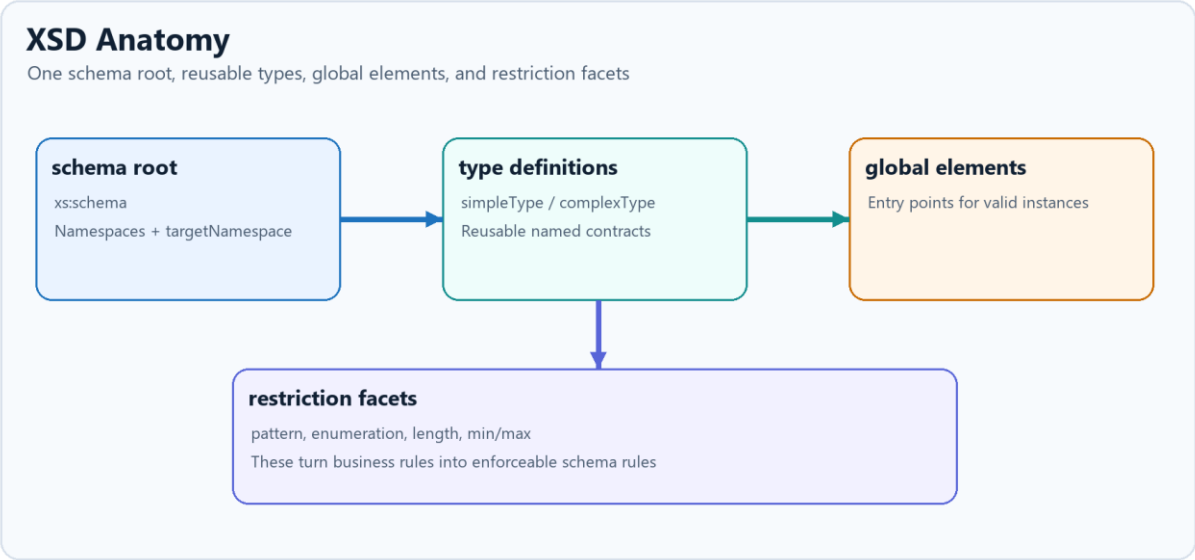
A) simpleType is only for comments, metadata blocks, and notes
B) simpleType can never use facets such as pattern, enumeration, or length
C) simpleType models scalar values; complexType models structured XML content

D) complexType is used only for attributes, comments, and annotations
E) There is no difference between them in practical schemas

98. Which facet should be chosen to validate an email-like text format in Week 3 style exercises?

A) enumeration
B) pattern
C) minInclusive
D) choice
E) all

99. In the Week 3 structure diagram, what sits between the schema root block and the global elements block?



A) A JSON serializer box
B) An HTTP controller layer
C) A CDATA processor
D) The type definitions block
E) The answer key

100. Why was Week 3 modeling guidance concerned with required and optional fields?

A) Because XSD cannot represent optional content in any form
B) Because only attributes may be optional in XML contracts
C) Because namespaces decide whether data is required or omitted
D) Because optionality exists only in DTD and never in XSD
E) Because contracts must show what is required and optional

101. What is one benefit of namespace-aware instance XML mentioned in the lecture?

A) It improves tooling interoperability
B) It disables validation events
C) It converts local elements into comments
D) It prevents all schemas from using facets
E) It removes the need for targetNamespace

102. When the same bad business value is tested against DTD and XSD, what outcome does Week 3 want students to notice?

A) DTD usually gives richer type-aware diagnostics than XSD does
B) Both systems always behave identically for every bad value

C) XSD can fail with more specific constraint errors
D) Neither DTD nor XSD can detect bad values in practice
E) Only JSON can report the error with meaningful detail

103. Which XSD type best matches a timestamp containing both date and time?

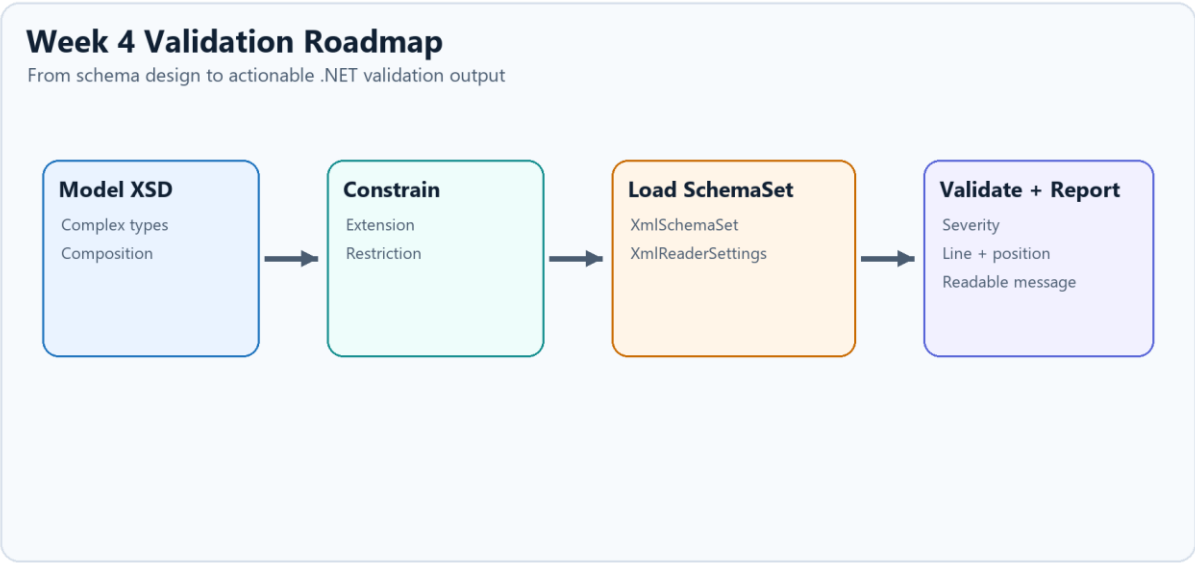
A) xs:boolean
B) xs:dateTime
C) xs:integer
D) xs:token
E) xs:ENTITIES

104. Which statement best matches the Week 3 evaluation focus for lab work?

A) Only short element names matter during grading
B) Only JSON export matters for the final score
C) Only browser rendering matters during schema review
D) Correctness, type reuse, and readable validation matter
E) Only visual styling matters in the submitted files

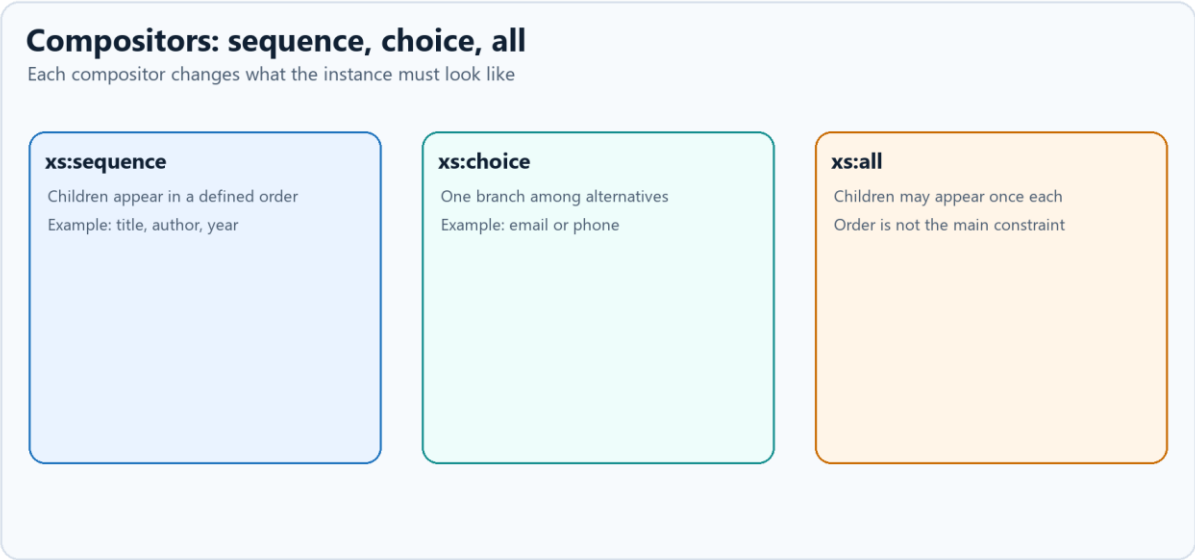
Week 4

105. In the Week 4 roadmap diagram, what is the final stage on the far right?



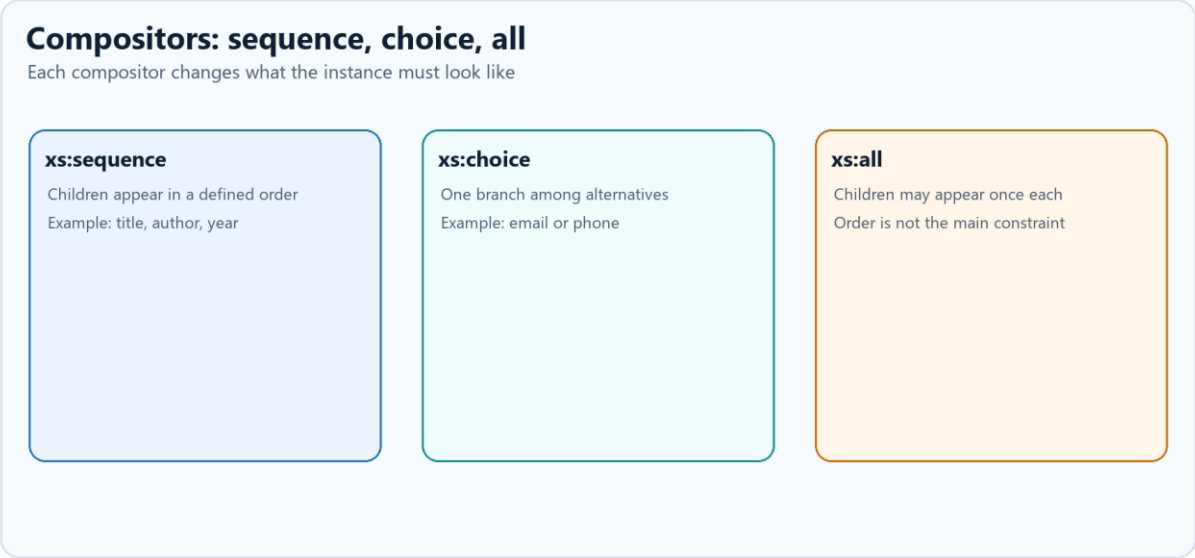
A) Model XSD
B) Constrain
C) Load SchemaSet
D) Compose JSON
E) Validate + Report

106. What does xs:choice mean in an XSD content model?



A) All children must appear in a fixed order
B) One of several child alternatives may appear
C) Every child may repeat without limits
D) No child elements are allowed
E) Only attributes may be validated

107. In the compositors diagram, what is the main characteristic of xs:all?



A) Its children may appear once each, and order is not central
B) It requires the same child to repeat forever in one branch
C) It works only with attributes and never with child elements
D) It always demands a namespace prefix on values and attributes
E) It converts elements into text nodes during validation

108. What does xs:sequence enforce?

A) A random order chosen by the parser at runtime
B) A substitution group with multiple concrete members
C) A defined left-to-right order of child elements
D) Only attribute validation with no child elements
E) Only text content without any structural checks

109. Which modeling situation is a good fit for xs:choice?

A) A record that must always contain firstName, lastName, and id
B) A list where every child must appear twice in fixed order
C) A schema with no child elements or alternatives
D) A contact element that may contain email or phone
E) A type that should ignore validation and business rules entirely

110. Why are named complex types useful in Week 4 schema design?

A) They make XML independent from schemas and validation rules
B) They are required only for comments and documentation blocks
C) They remove the need for namespaces in shared contracts
D) They force every element to be optional inside the model
E) They support reuse and keep shared structure consistent

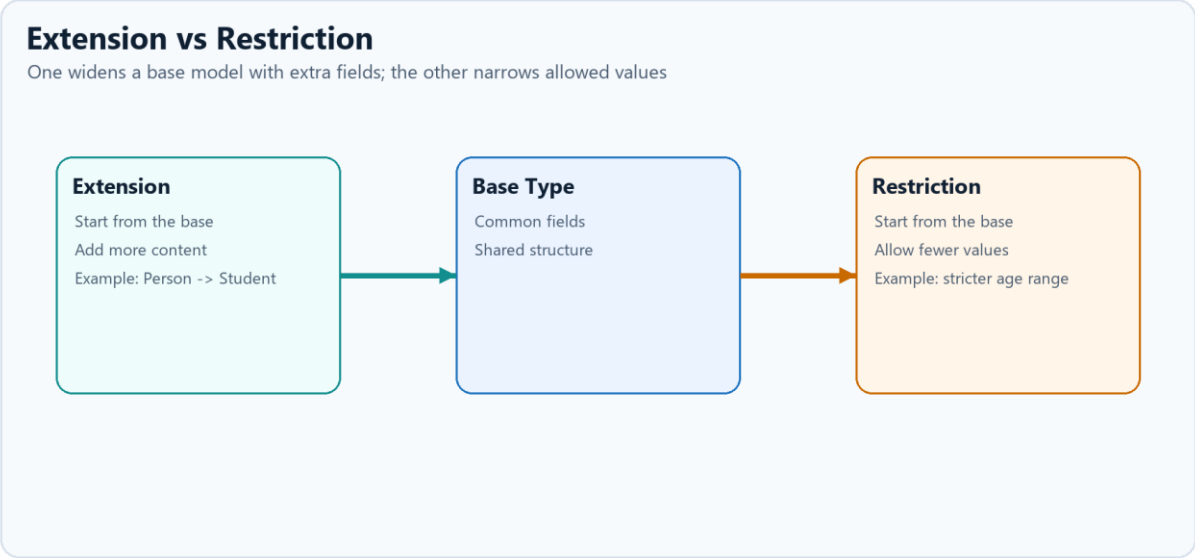
111. Which example best reflects the lecture's idea of nested complex types?

A) A decimal field stored as a comment near the root element
B) A University with nested Department and Student elements
C) A single attribute with no hierarchy or child structure
D) A JSON array inside an XML comment for display only
E) A root element with no children or nested content

112. What is meant by constraint layering in Week 4 discussions?

A) Applying structure and value rules across related nested types
B) Stacking multiple XML files into one commented wrapper
C) Removing all restrictions from child elements and attributes
D) Using only xs:any for every node in the schema
E) Turning elements into binary values before validation

113. In the extension-versus-restriction diagram, what does xs:extension do in a complexContent design?

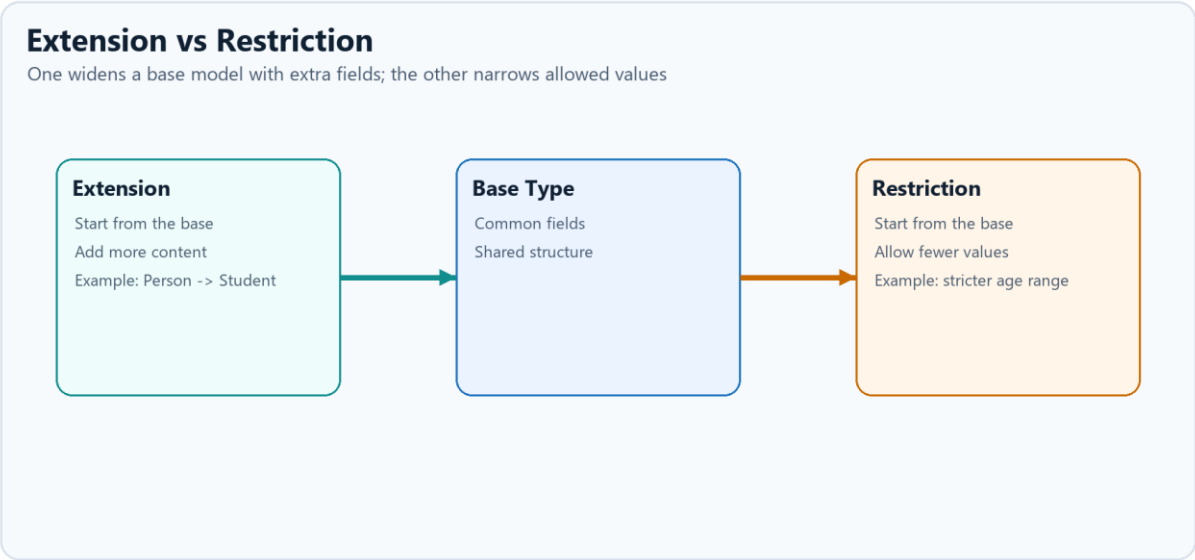


A) It deletes the base type
B) It changes XML into JSON
C) It adds content to a base type
D) It disables validation messages
E) It replaces namespaces with attributes

114. What does xs:restriction generally do?

A) It makes the type broader and less controlled
B) It adds unrelated child trees from another schema
C) It removes the base type from the document
D) It narrows the allowed values or structure of a type
E) It turns element content into comments

115. Which scenario matches extension better than restriction?

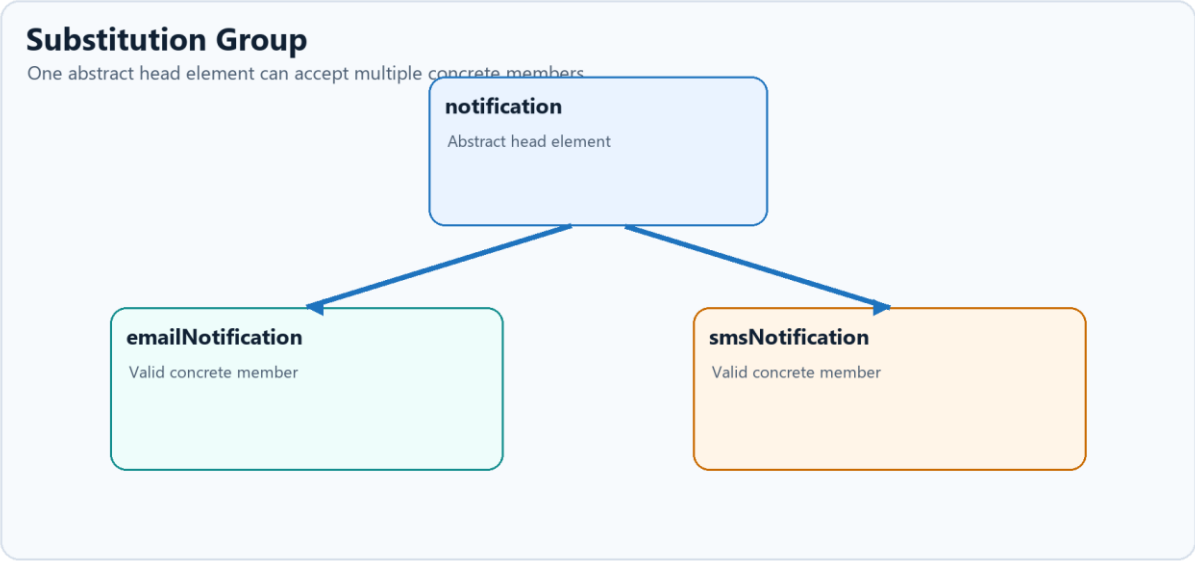


A) Forbidding a value that used to be valid
B) Reducing a maximum length from 20 to 10
C) Removing several child elements from a model
D) Making a numeric range narrower
E) Adding Student fields to a Person base type

116. Which scenario matches restriction best?

A) Adding an optional note field to a base type
B) Tightening a rule so fewer values remain valid
C) Creating a reusable namespace alias
D) Turning every child into a substitution group member
E) Skipping schema validation in code

117. In the substitution group diagram, what problem do substitution groups solve?

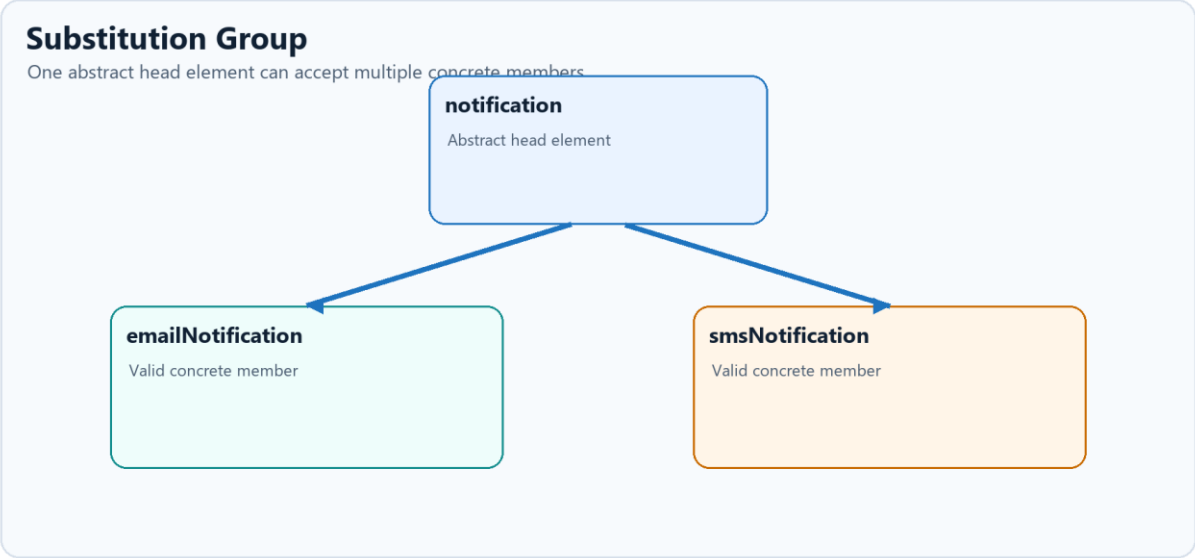


A) They let one head element accept several concrete members
B) They eliminate the need for global elements in every schema
C) They disable validation errors for unknown elements entirely

D) They replace all attributes with child nodes by default

E) They are used only in DTD and never in XSD

118. In the substitution group diagram, which element is shown as the abstract head?



A) message

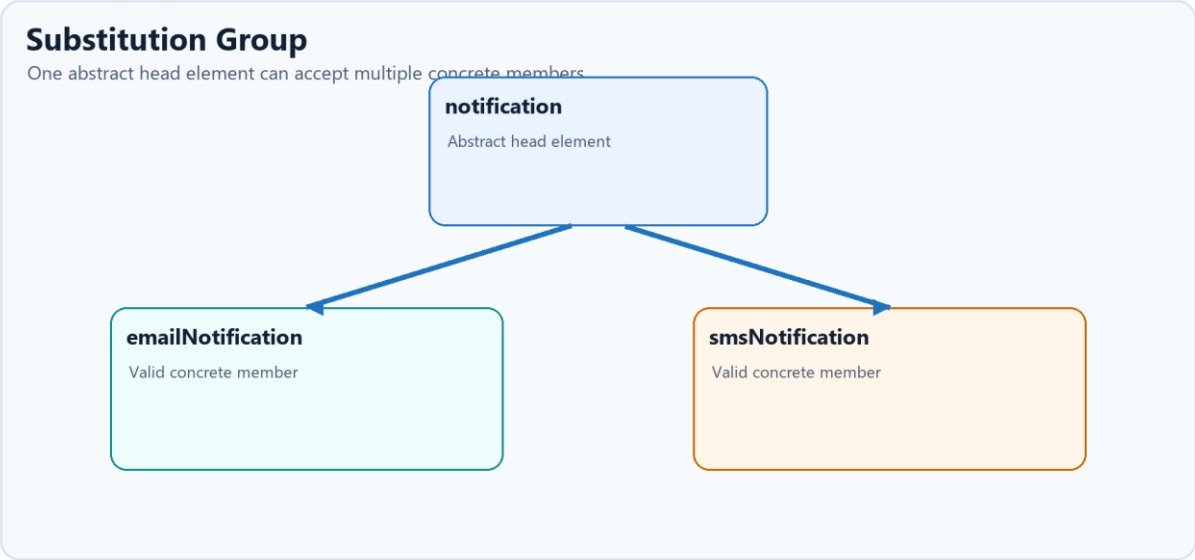
B) alert

C) notification

D) emailNotification

E) smsMessage

119. Which element name in the lecture diagram is a member that can substitute for the head element?



A) schemaRoot

B) validatorOutput

C) targetNamespace

D) emailNotification

E) XmlSchemaSet

120. Which .NET type is used to hold one or more XSD schemas before validation?

A) XmlDocument
B) XmlNamespaceManager
C) XPathNavigator
D) XmlSerializer
E) XmlSchemaSet

121. Which settings object is commonly configured when validating XML with an XmlReader-based pipeline?

A) StringBuilder
B) XmlReaderSettings
C) UriBuilder
D) CultureInfo
E) HttpClientHandler

122. What behavior was recommended for the Week 4 validator tool?

A) Capture all validation problems
B) Stop after reading the root element only
C) Convert errors into comments silently
D) Ignore line numbers to reduce output size
E) Validate only attributes and skip elements

123. Which information combination best matches the lecture's idea of a good validation report?

A) Only the schema file name and folder path
B) Only the current machine name and user account
C) Severity, line info, and a human-readable message
D) Only a boolean true or false with no explanation
E) Only the total node count in the current document

124. Why are line number and line position valuable in validator output?

A) They replace the need for validation messages entirely
B) They show the HTTP status code of the API
C) They tell the program which JSON serializer to use
D) They help you locate and fix the failing XML content
E) They allow namespaces to be omitted

125. Which item appears in the Week 4 minimum requirements?

A) A complete replacement of XML with CSV files
B) An XSLT text exporter instead of schema work
C) A machine-learning classifier for element names
D) A browser-only validator with no C# code
E) At least two named complex types in the schema design

126. Which design expectation was explicitly listed for the lab?

A) Avoid all compositors and use only text nodes
B) Use one extension rule and one restriction rule
C) Use only untyped elements without any schema
D) Skip error reporting because the XML is small
E) Remove namespaces from the project

127. Which scenario was specifically expected as part of the Week 4 schema work?

A) Show one substitution group case
B) Replace XML with a relational table dump
C) Use only comments as documentation and no validation

D) Avoid reusable types to keep the schema short
E) Disable namespaces permanently

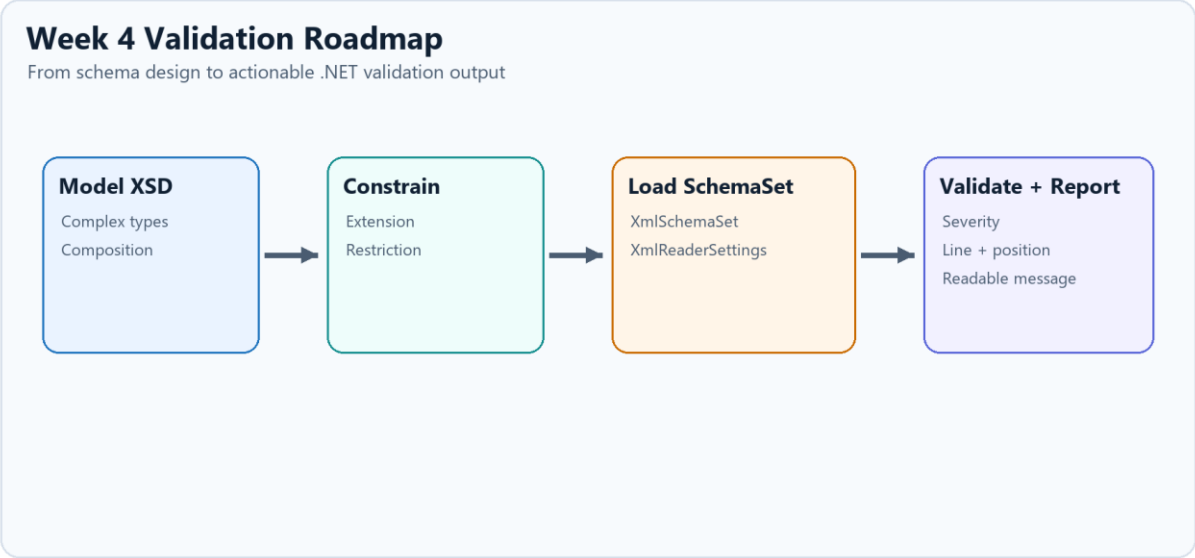
128. If an XSD uses xs:sequence and the instance swaps the child order, what should happen?

A) The parser silently sorts the children
B) The instance becomes valid if the names still exist
C) Validation should fail because the order was violated
D) The schema converts itself to xs:all
E) Only attributes are checked, so order never matters

129. In the Week 4 substitution group example, what happens if the payload uses an unknown element that is not a valid member?

A) The validator promotes it to the nearest known member automatically
B) The element is treated as a harmless comment and ignored
C) The payload becomes a DTD instance automatically during parsing
D) Validation fails because the element is not a valid member
E) The namespace manager adds it silently to the allowed group

130. In the roadmap diagram, which stage explicitly mentions XmlSchemaSet and XmlReaderSettings?



A) Model XSD
B) Constrain
C) Validate + Report
D) Data Binding
E) Load SchemaSet

131. Which statement distinguishes xs:choice from xs:sequence most accurately?

A) choice enforces exact order while sequence allows alternatives
B) choice picks alternatives; sequence enforces order
C) choice is only for attributes, while sequence is only for text
D) choice disables validation, while sequence enables it
E) choice and sequence are identical in XSD

132. Which statement about xs:all is correct?

A) Each child may appear once, and order is not central
B) It forces one branch among several alternatives in a choice model

C) It always allows unlimited repetition of every child element
D) It can validate only attributes and never child elements
E) It is a DTD-only feature with no XSD equivalent

133. Which mechanism preserves the base type and adds more content to it?

A) restriction
B) choice
C) extension
D) all
E) enumeration

134. Why are nested complex types useful in real XML contracts?

A) They remove the need for root elements in large documents
B) They stop validation from using namespaces and schemas
C) They make every element optional regardless of business rules
D) They exist only for styling XML in browsers and editors
E) They model real hierarchies such as university and department

135. After the schemas are loaded into the validation pipeline, what major task follows next?

A) Convert the XSD into JSON Schema
B) Validate the XML and report problems
C) Render HTML from the schema
D) Replace all types with xs:string
E) Disable reader settings

136. Which item belongs in an actionable validation message according to the lecture?

A) A human-readable explanation of what failed
B) The weather at the time of validation
C) The browser bookmarks of the student
D) The USB device list of the computer
E) A random GUID with no context

137. When are substitution groups especially helpful?

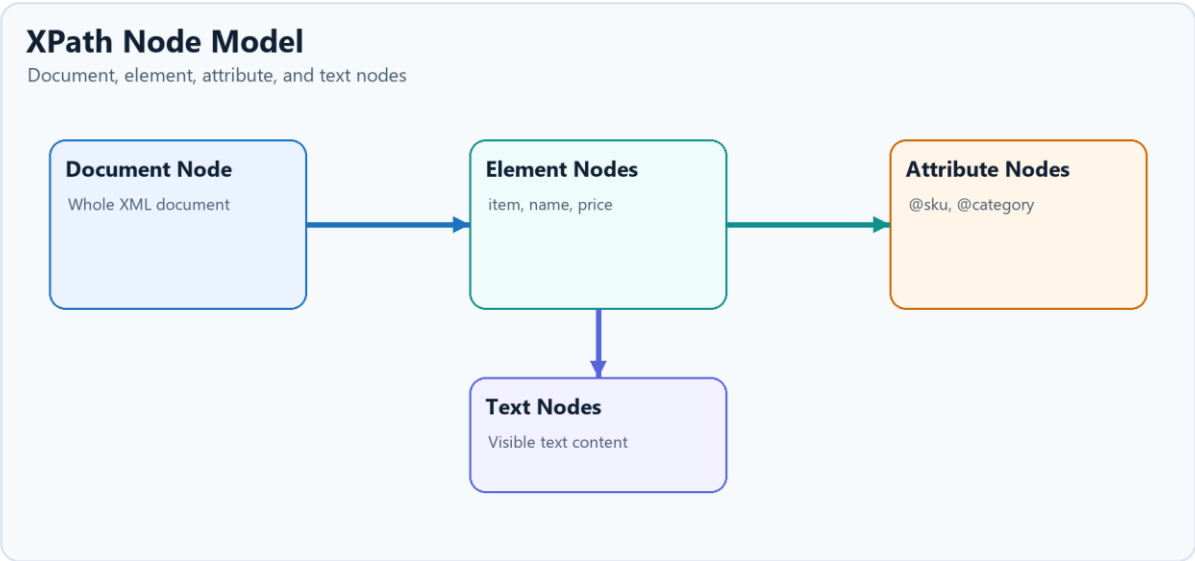
A) When all elements must always be identical in content and structure
B) When schemas should reject every variant payload by design
C) When one slot expects a common head but payloads may vary by concrete type
D) When XML must be converted into plain text only for export
E) When namespaces need to be removed from the schema permanently

138. Which statement best summarizes the Week 4 lecture focus?

A) Only JSON serialization and HTTP routing were studied in detail
B) Only XPath filtering and counting were studied this week
C) Only XSD namespace prefixes and URI syntax were studied
D) It focused on complex types, inheritance, and validator quality
E) Only XSLT grouping and sorting were studied this week

Week 5

139. In the Week 5 node model diagram, the labels @sku and @category are shown as which node type?



A) Document nodes
B) Text nodes
C) Comment nodes
D) Attribute nodes
E) Namespace declarations only

140. What does the XPath expression /catalog/item represent?

XPath Core Syntax

The symbols students use in most Week 5 queries

Syntax	Meaning	Example
/	Absolute path	/catalog/item
//	Descendant search	//price
@	Attribute selection	//item/@sku
[...]	Predicate filter	//item[price > 500]
text()	Text child nodes	//name/text()

A) An absolute path starting from the document root
B) A descendant search that can match nodes at any depth
C) An attribute lookup against the current context node
D) A text-node conversion applied to a selected element
E) A namespace declaration or prefix binding statement

141. In the XPath core syntax diagram, what does the double slash in //price mean?

XPath Core Syntax

The symbols students use in most Week 5 queries

Syntax	Meaning	Example
/	Absolute path	/catalog/item
//	Descendant search	//price
@	Attribute selection	//item/@sku
[...]	Predicate filter	//item[price > 500]
text()	Text child nodes	//name/text()

A) Only direct child elements named price are matched
B) Only attribute nodes named price are matched anywhere
C) The result is always treated as plain text content
D) The query becomes invalid in standard .NET XPath engines
E) Search for descendant nodes named price anywhere in the document

142. What does the @ symbol mean in an XPath expression such as //item/@sku?

A) It means descendant-or-self
B) It selects an attribute node
C) It selects a text node only
D) It starts a comment filter
E) It declares a namespace prefix

143. In //item[price > 500], what is [price > 500]?

A) A namespace condition
B) A text node converter
C) A predicate filter
D) A schema declaration
E) A processing instruction

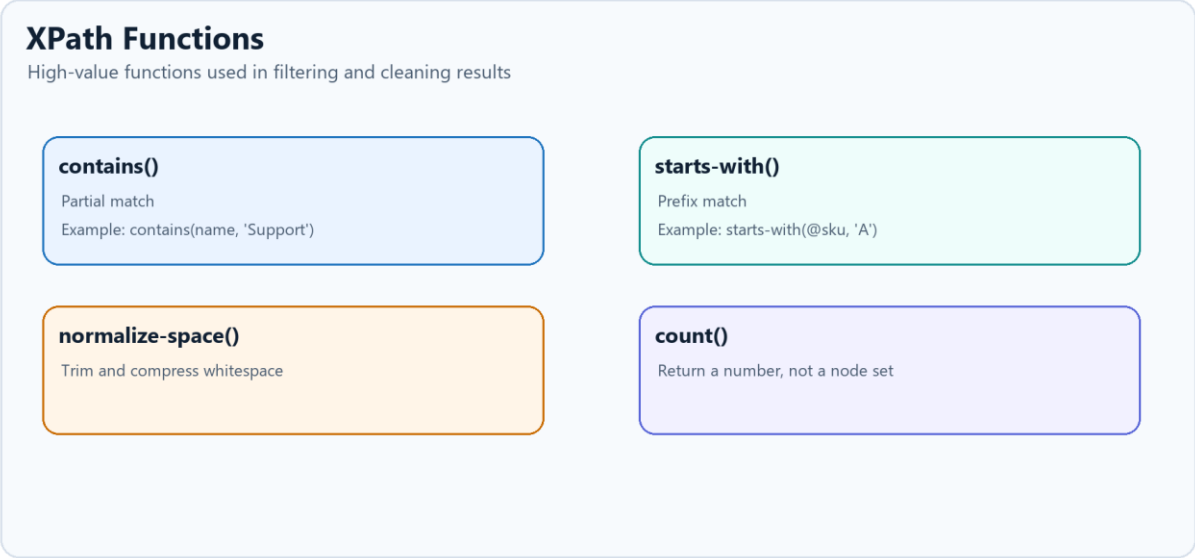
144. What does the expression //item[price > 500] return?

A) Only the numeric price values from matching nodes
B) Only the root element of the XML document
C) Only the attribute nodes attached to matching items
D) The whole item nodes that satisfy the predicate
E) A Boolean true or false value only

145. What does text() select in XPath?

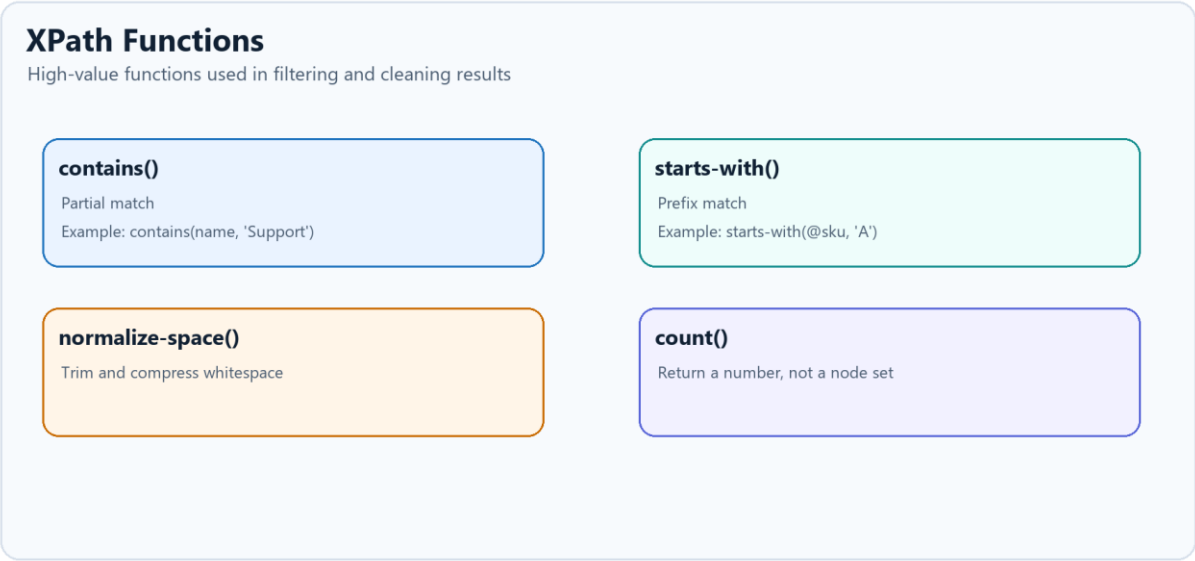
A) The text child nodes of the current element selection
B) All comment nodes in the current XML document
C) Only attribute values from the current element set
D) Only namespace URIs declared in the document
E) Only XML declarations and processing instructions

146. Which function is appropriate for finding names that contain a word such as Support anywhere in the text?



A) count()
B) normalize-space()
C) last()
D) position()
E) contains()

147. In the XPath functions diagram, what does starts-with(@sku,'A') check?



A) Whether the item has no sku attribute
B) Whether the sku attribute value begins with A
C) Whether the sku attribute ends with A
D) Whether the item is the first child
E) Whether the attribute is numeric

148. What does normalize-space() do?

A) It converts XML to JSON before XPath evaluation
B) It counts descendant nodes in the selected subtree
C) It trims and compresses whitespace in a string value

D) It adds indentation and visual spacing for display
E) It forces a namespace prefix onto element names

149. What kind of result does count(/catalog/item) produce?

A) A node set of all matching items
B) An attribute list
C) A namespace map
D) A numeric count value
E) A copy of the XML tree

150. Which statement about / is correct in XPath?

A) It marks an absolute path from the root
B) It means any descendant at any depth
C) It selects attribute nodes only
D) It is used only inside functions
E) It disables predicates

151. What would XPath try to find if the @ symbol were removed from //item/@sku?

A) A namespace named sku
B) A processing instruction named sku
C) A comment node named sku
D) A schema type named sku
E) A child element named sku under item

152. What should a student remember about most XPath string matching in these examples?

A) It is always case-insensitive in every XPath engine
B) It is case-sensitive unless the logic explicitly handles casing
C) It always sorts string matches alphabetically before returning
D) It ignores namespaces completely during matching
E) It works only for attribute values and never element text

153. What does /catalog/item[2] select?

A) Every second item anywhere in the document
B) The second attribute of catalog
C) The second item child under catalog
D) All items except the first
E) Only items with id=2

154. What does or mean inside an XPath predicate?

A) Both conditions must always be true
B) The expression switches to arithmetic mode
C) Only attributes may be tested
D) Either condition being true is enough for a match
E) The result becomes text automatically

155. What does not(supplier/email) detect?

A) A missing supplier/email path
B) An email node with uppercase letters
C) Only emails longer than 20 characters
D) A namespace mismatch on the root
E) A duplicate catalog item

156. What is the benefit of not(normalize-space(supplier/email))?

A) It turns email text into an attribute
B) It counts how many email nodes exist
C) It forces every email into lowercase
D) It strips namespaces from the email path
E) It catches blank or whitespace-only email values

157. What is usually required for namespace-aware XPath queries in .NET?

A) A DTD declared inside the XPath string or query context
B) A registered namespace prefix bound to the correct URI
C) A JSON formatter for the XML tree and navigator
D) A serializer generated from XSLT and attached at runtime
E) A second root element inside the same XML document

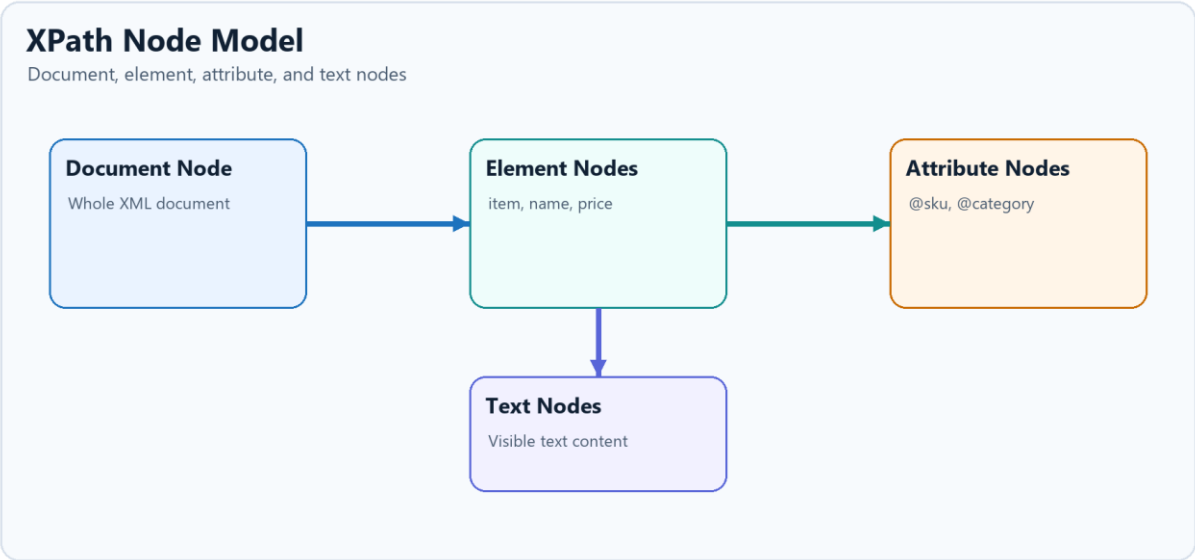
158. Why is an indexed path like //*[@id='deck']/div[4]/section[1] considered brittle?

A) Because XPath cannot use square brackets in indexed paths
B) Because it ignores all attribute filters and predicates
C) Because order changes can break it
D) Because it can match only text nodes and not elements
E) Because it forces XML to become invalid during parsing

159. Which statement matches the Week 5 lab expectation?

A) Students should avoid writing more than two queries
B) Students should build an XSD validator instead of XPath queries
C) Students should focus only on HTML rendering
D) Students should write 12 to 20 XPath queries on the dataset
E) Students should replace XML with CSV

160. Which node type represents the whole XML document in the Week 5 diagram?



A) Document node
B) Attribute node
C) Text node
D) Index node
E) Comment node

161. Why might someone use string(name) before applying a match function?

A) To add a namespace automatically to the selected element
B) To switch the query to schema-validation mode first
C) To remove all child nodes from the XML result tree
D) To force a numeric comparison instead of a string test
E) To turn the selected child element into a string before matching

162. What does the dot . usually refer to in XPath predicates and functions?

A) The root document only
B) The current node context
C) Any attribute in the document
D) The previous sibling always
E) A fixed namespace alias

163. Why is normalize-space(.) useful when checking text values?

A) It converts text into an XSD type before comparison
B) It orders sibling elements by name before filtering
C) It trims and compresses the current node's text value
D) It loads external namespaces from the web automatically
E) It changes element names to lowercase for matching

164. Which expression style is appropriate when the goal is to return the number of hardware items, not the nodes themselves?

A) Use a plain path with no function at all
B) Use starts-with() on the root element
C) Use text() against the entire catalog
D) Use count(...) around a filtered item path
E) Use normalize-space() on each attribute

165. Why are missing-field and threshold queries useful beyond simple data retrieval?

A) They support data-quality checks as well as ordinary retrieval
B) They force XML to become valid XSD before querying
C) They can only be used inside XSLT transformations
D) They replace the need for namespaces in XML queries
E) They always return a single root node as the result

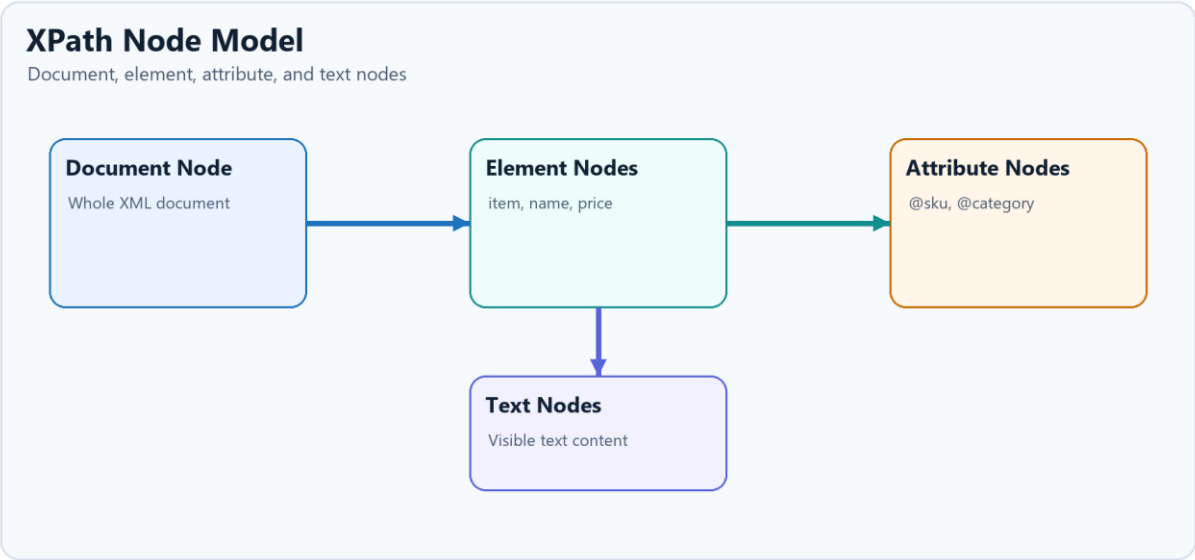
166. Which expression form best represents 'find item nodes whose price is greater than 500'?

A) /catalog/@price > 500
B) //price[item > 500]
C) count(//item/price > 500)
D) //item/@price > 500
E) //item[price > 500]

167. What is the difference between //name and //name/text()?

A) There is no difference at all between the two expressions
B) The first selects name elements; the second selects their text nodes
C) The first selects attributes, while the second selects comments
D) The first is JSON-style, while the second is XML-style
E) The second can be used only in schemas and validators

168. In the node model diagram, which node type is shown below the element nodes box?



A) Document nodes
B) Attribute nodes
C) Predicate nodes
D) Text nodes
E) Schema nodes

169. What kind of results are produced by //item/@sku in the lecture examples?

A) The matching @sku attribute nodes selected from item elements
B) Whole item subtrees only, without separate attributes
C) The document root element and nothing below it
D) Only comments and text nodes from inside each item
E) Only namespace declarations attached to the root

170. Which advanced skill from Week 5 extends beyond basic absolute paths and simple filters?

A) Removing every namespace from the document
B) Replacing XPath with SQL
C) Turning XML into CSV manually
D) Ignoring missing data cases
E) Writing namespace-aware XPath in .NET

171. Why should students be cautious with heavily indexed XPath paths?

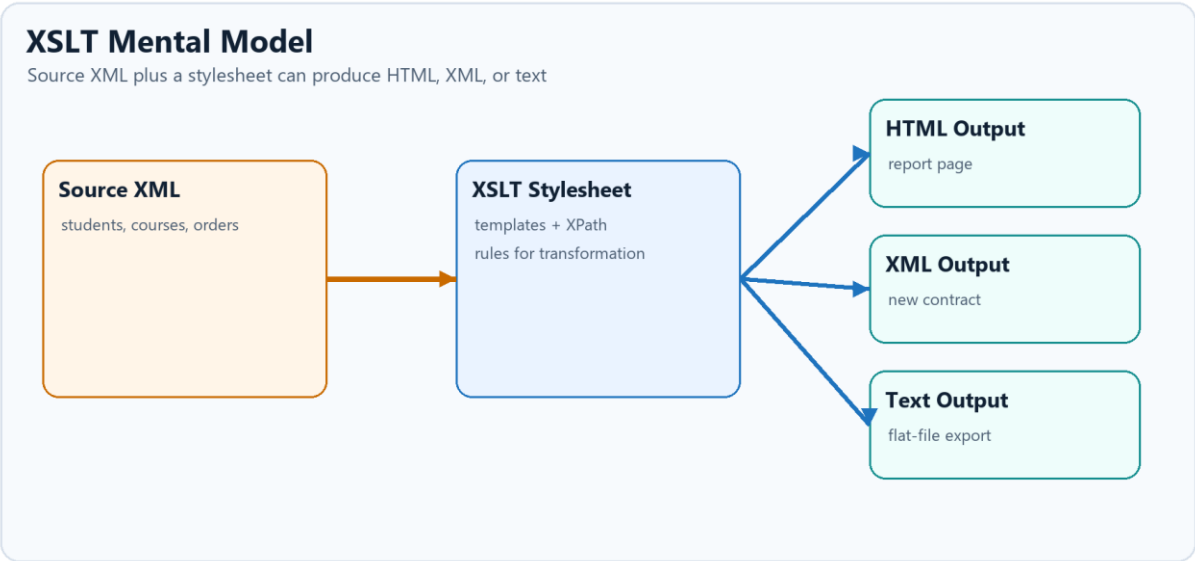
A) Indexes are forbidden in XPath expressions and predicates
B) They depend on document order and break when structure shifts
C) They always return attributes instead of matching elements
D) They cannot be used in .NET XPath implementations
E) They disable all predicates and namespace matching

172. Which statement best summarizes the main Week 5 learning outcome?

A) Build SOAP services with WSDL contracts and service bindings
B) Create XSLT reports and text exports from source XML
C) Write XPath queries in .NET to filter and count XML data
D) Stream large XML files with XmlReader and subtree skipping
E) Model inheritance with XSD extension and restriction

Week 6

173. In the Week 6 mental model diagram, what sits between the source XML and the produced outputs?



A) An XmlReader cursor that scans source nodes one by one
B) A JSON serializer that rewrites the XML into objects
C) An XSLT stylesheet containing template rules
D) A DTD external subset used only for structural validation
E) A namespace manager used only for prefix registration

174. Which language does XSLT rely on for matching, selecting, and testing nodes?

A) SQL
B) DTD
C) CSS
D) WSDL
E) XPath

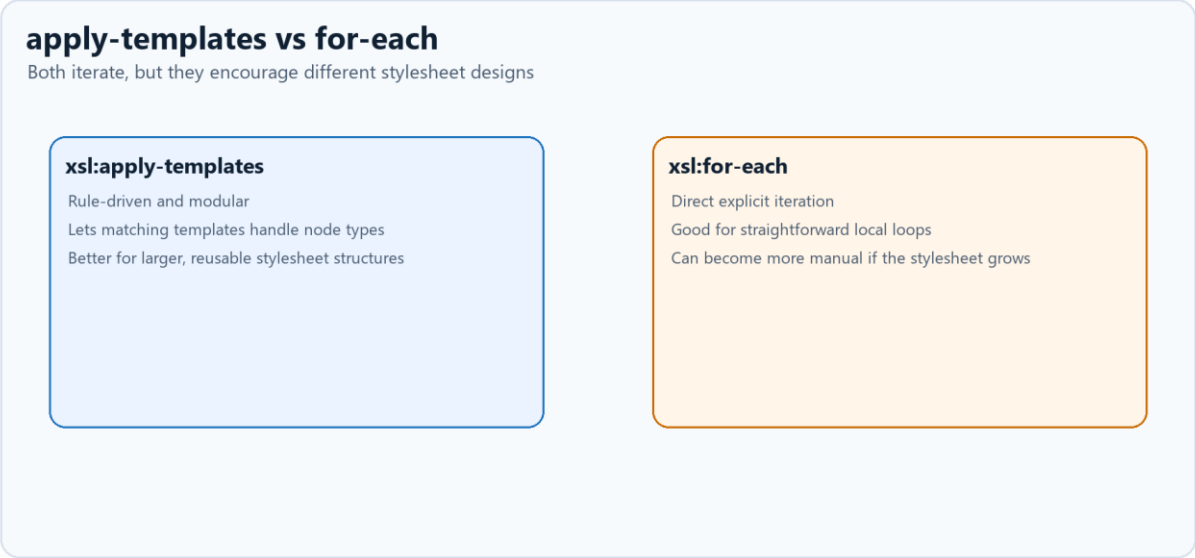
175. What key idea about XSLT processing was emphasized in the lecture?

A) It is template-driven: the processor picks matching templates
B) It always runs line-by-line like a shell script
C) It ignores XML structure entirely during transformation
D) It can generate only HTML and not XML or text
E) It works only without XPath expressions or match rules

176. What does an xsl:template with a match attribute define?

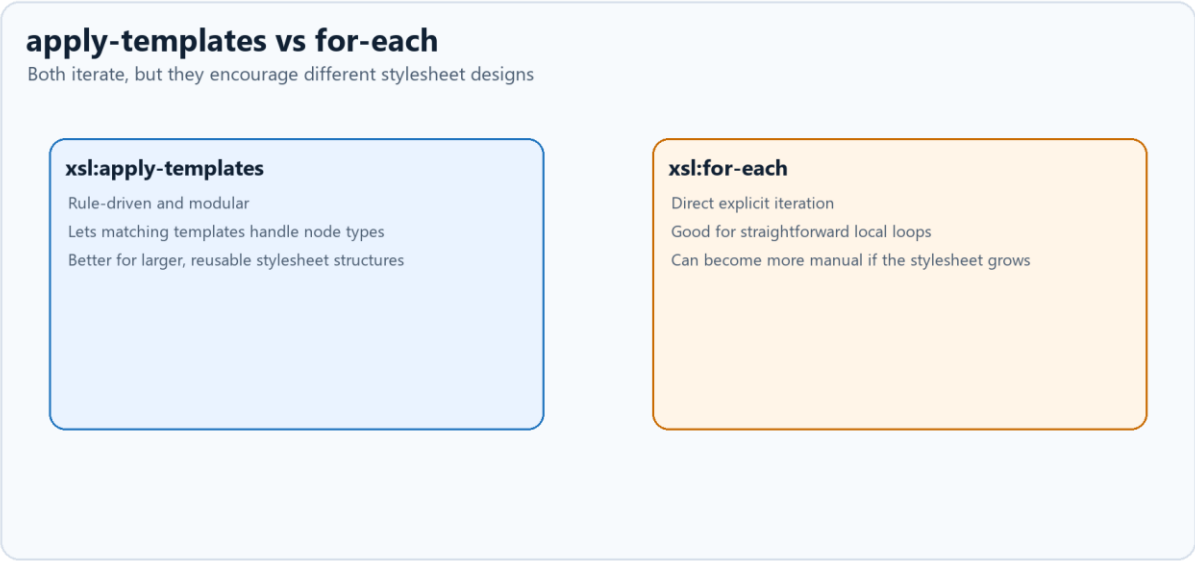
A) A database transaction that updates the XML source
B) A namespace alias used for serializer configuration
C) A numeric restriction inside an XSD-like contract
D) A transformation rule for nodes that match the pattern
E) A fixed output file path for the transformation result

177. Which construct is more aligned with modular, template-driven processing?



A) xsl:value-of only
B) xsl:apply-templates
C) xsl:output only
D) xsl:sort only
E) xsl:text only

178. In the apply-templates-versus-for-each diagram, when is xsl:for-each often a good fit?



A) When validation should be skipped for the XML input
B) When namespaces must be removed from the source tree
C) When you need explicit iteration over a selected node set
D) When only comments should be transformed into output
E) When no XPath is available inside the stylesheet

179. What is xsl:if mainly used for?

A) Sorting output alphabetically by one selected field
B) Declaring namespaces for the stylesheet and output
C) Grouping nodes by key across the whole document

D) Compiling the stylesheet in .NET before execution
E) Handling one conditional branch when a test is true

180. Which XSLT construct is most suitable for multi-branch logic?

A) xsl:choose with xsl:when and xsl:otherwise
B) xsl:attribute-set used alone without branch conditions
C) xsl:comment inserted into the output document
D) xsl:preserve-space applied to source element names
E) xsl:key used alone without any conditional structure

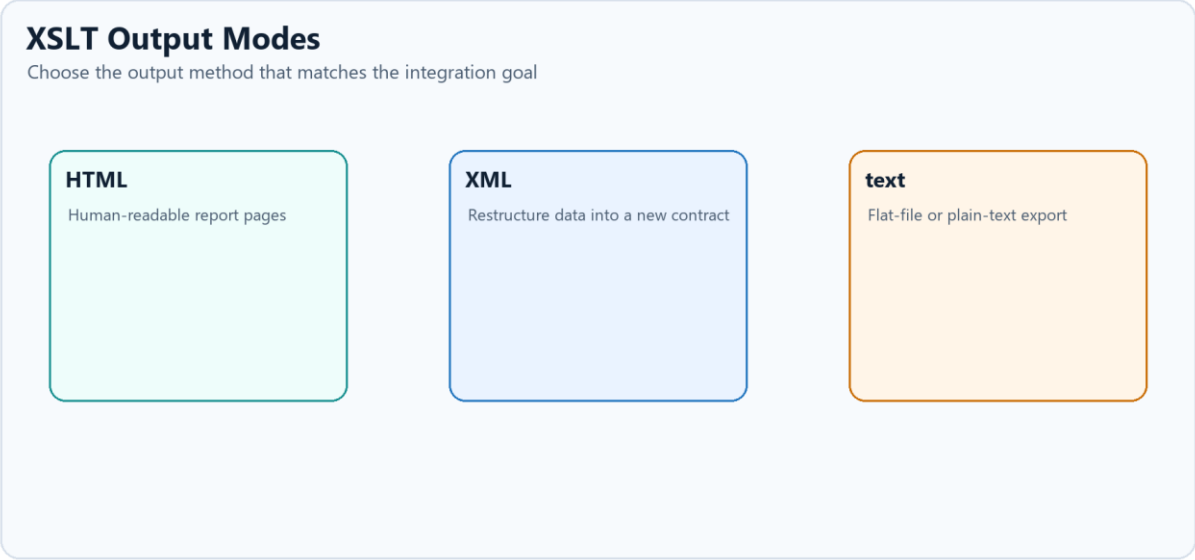
181. Where can xsl:sort be placed in the patterns discussed during Week 6?

A) Only inside xsl:stylesheet attributes
B) Only inside xsl:output
C) Only inside named templates with no selection
D) Inside apply-templates or for-each
E) Only inside XML comments

182. How is grouping commonly approached in XSLT 1.0 according to the lecture?

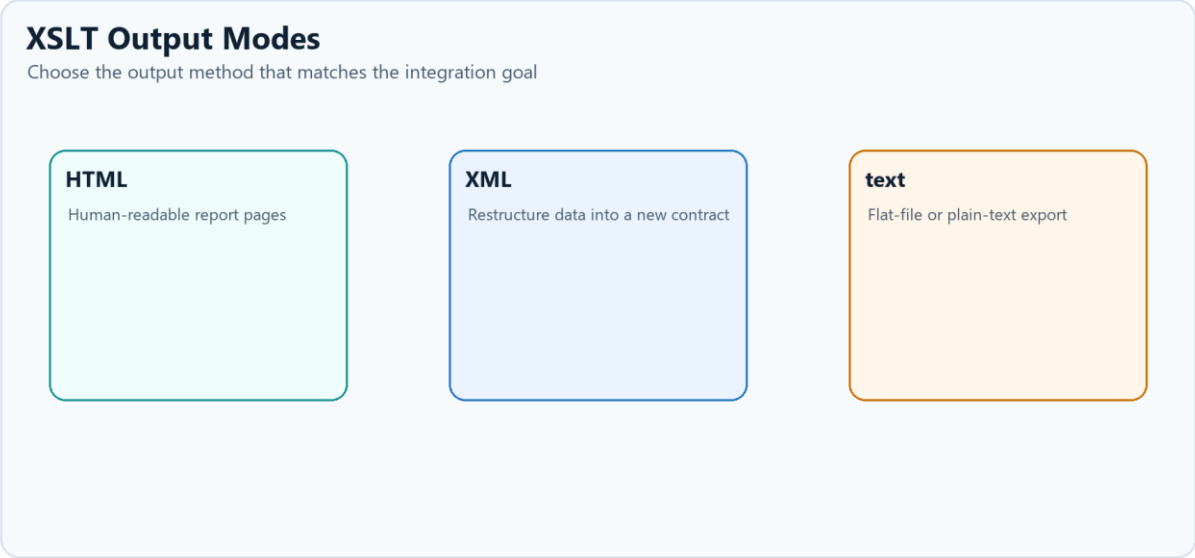
A) With SQL joins against the XML file
B) With xsl:key and leader-node style logic
C) By removing all duplicate nodes before transform
D) By disabling namespaces in the processor
E) By converting XML to JSON first

183. Which output mode is appropriate when producing a browser-friendly report page?



A) Binary
B) CDATA
C) HTML
D) DTD
E) Schema

184. In the XSLT output modes diagram, which output mode should be used for a plain text export file?



A) HTML
B) JSON
C) SOAP
D) CDATA
E) text

185. Why are parameters useful in an XSLT stylesheet?

A) They let one stylesheet be reused with different input values
B) They delete unmatched nodes automatically during output
C) They replace XPath selections completely across the stylesheet
D) They disable output methods such as HTML, XML, or text
E) They force all values to be strings at every call site

186. How is a named template invoked in XSLT?

A) With xsl:apply-imports
B) With xsl:result-document
C) With xsl:analyze-string
D) With xsl:call-template
E) With xsl:fallback

187. Which instruction is used to pass an argument into a called template?

A) xsl:with-output
B) xsl:with-param
C) xsl:key
D) xsl:import
E) xsl:comment

188. Which .NET class is used in the lecture for running compiled XSLT transforms?

A) XmlSchemaSet
B) XmlDocument
C) XslCompiledTransform
D) XPathNavigatorBuilder
E) XmlSerializerNamespaces

189. Which .NET helper object is used to pass external parameters into an XSLT transformation?

A) XmlReaderSettings
B) XmlNamespaceManager
C) XmlSchemaInference
D) StringBuilder
E) XsltArgumentList

190. What was the goal of the first major Week 6 example?

A) Transform XML into an HTML report
B) Validate XML against a DTD
C) Build an XPath-only counter
D) Stream a feed with XmlReader
E) Generate a SOAP WSDL

191. What was the goal of the legacy-to-new-contract example?

A) To remove every element from the source
B) To test only whitespace behavior
C) To build a schema validator
D) To reshape old XML into a new XML contract
E) To create a binary export format

192. Why is xsl:apply-templates often preferred in larger stylesheets?

A) It disables XPath for better performance in large stylesheets
B) It encourages modular template matching instead of one large loop
C) It can output only text files rather than XML or HTML
D) It removes the need for root templates and entry matches
E) It automatically validates XSD before transformation begins

193. Which misunderstanding was explicitly warned against in Week 6?

A) Thinking that XML can never contain attributes or metadata
B) Thinking that XSLT cannot produce XML output documents
C) Thinking XSLT works line by line instead of by template rules
D) Thinking that HTML output makes XPath unnecessary
E) Thinking that namespaces are impossible in XML documents

194. Which set correctly lists the output modes emphasized in Week 6?

A) CSV, YAML, and TOML
B) Binary, JSON, and PDF
C) DTD, XSD, and SOAP
D) PNG, SVG, and JPEG
E) HTML, XML, and text

195. Why are named templates useful beyond the immediate current node match?

A) They package reusable logic callable from multiple places
B) They replace the stylesheet root element entirely
C) They remove the need for output settings and methods
D) They force the transform to produce JSON-like output
E) They turn every node into text automatically at runtime

196. Which statement best distinguishes xsl:choose from xsl:if?

A) choose is for namespaces, while if is for sorting
B) choose always removes unmatched nodes from the output
C) if is only for attributes, choose is only for elements

D) choose handles multiple branches, while if handles one condition
E) There is no meaningful difference between the two

197. If a student is asked to sort courses by available seats in an XSLT task, which instruction is the most relevant?

A) xsl:key
B) xsl:sort
C) xsl:comment
D) xsl:number-format
E) xsl:namespace-alias

198. What does the Week 6 mental model make clear about XSLT output?

A) It can only produce HTML pages for browser viewing
B) It can only produce XML trees and never text output
C) It can produce HTML, XML, or text depending on the stylesheet
D) It cannot generate plain text for flat-file exports
E) It always returns the original source document unchanged

199. Who decides which template runs for a matched node during an XSLT transformation?

A) The student manually selects each template at runtime
B) The namespace URI decides without any XPath
C) The output method decides
D) The source XML filename decides
E) The XSLT processor, based on matching rules

200. Where is XPath used inside XSLT according to the lecture?

A) In match patterns, select expressions, tests, and sorts
B) Only in the XML declaration line at the top
C) Only in output method attributes for HTML and text
D) Only in comments added for stylesheet documentation
E) Nowhere, because XSLT does not depend on XPath

201. Why is a parameterized text export valuable in practice?

A) It prevents any reuse of the stylesheet across different runs
B) It forces all nodes to be grouped before transformation
C) It removes the need for XPath predicates and tests
D) It lets one stylesheet support different filters
E) It works only for invalid XML and debugging samples

202. Which output method is most appropriate for a legacy flat-file style export?

A) HTML
B) XML
C) text
D) Schema
E) Binary

203. What is a practical expectation when running XSLT in .NET correctly?

A) The transform runs only if the XML has no namespaces
B) The output must always be HTML regardless of settings
C) Parameters cannot be passed from C# into the stylesheet
D) XslCompiledTransform is only for validation, not transformation
E) The output should go to a file, stream, or writer

204. What was the purpose of comparing apply-templates and for-each in practice?

A) To compare two transformation styles for clarity and structure
B) To prove that XPath is unnecessary in XSLT work
C) To remove templates from XSLT permanently after one run
D) To force every stylesheet into text mode for export
E) To avoid selecting nodes anywhere in the stylesheet

205. Which deliverable set reflects the required output variety in Week 6?

A) Only one HTML file and nothing else for submission
B) Only a schema and no transform output at all
C) Only console screenshots with no generated artifacts
D) Transform work that includes HTML, XML, and text outputs
E) Only JSON API responses from a web service demo

206. Which statement best summarizes Week 6?

A) The week focused on DTD attribute declarations and ID rules
B) It focused on XSLT as a template-driven XML language using XPath
C) The week focused on only XmlReader streaming and skipping
D) The week focused on SQL-only queries over XML datasets
E) The week focused on WSDL publishing only for SOAP services

Week 7

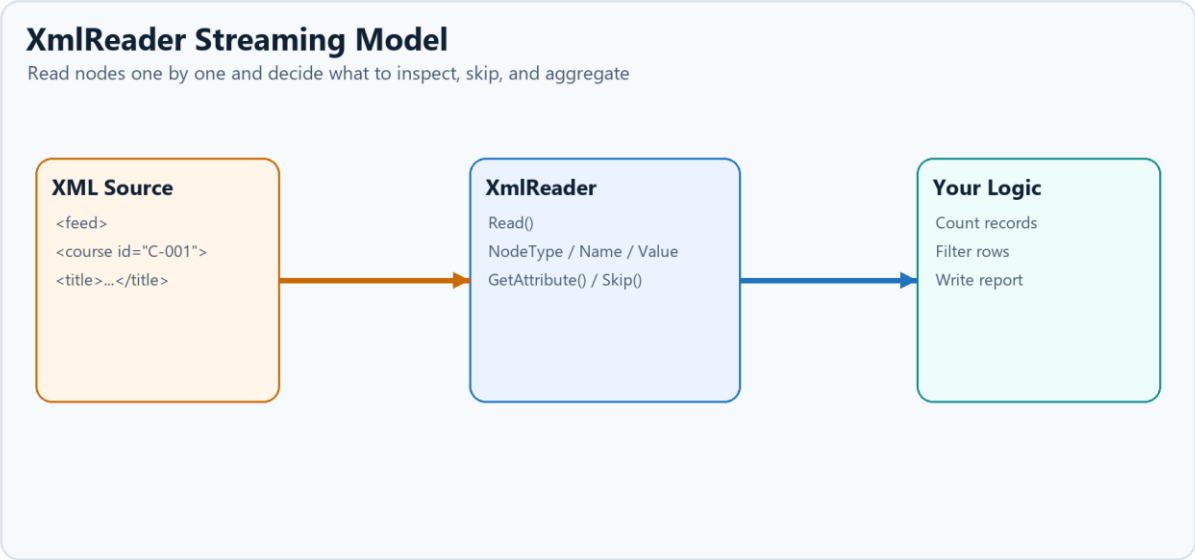
207. Which API is the best fit for very large XML feeds when low memory usage matters most?

A) XmlDocument with full tree loading
B) XmlReader with streaming access
C) An XSLT stylesheet alone
D) A DTD external subset
E) A namespace alias file

208. What is the correct mental model for XmlReader in Week 7?

A) It edits the XML tree in place like a DOM
B) It automatically aggregates totals without code
C) It converts XML directly to JSON
D) It is a forward-only reader cursor
E) It validates only attributes

209. In the streaming model diagram, which component sits between the XML source and your custom logic?



A) XmlSchemaSet
B) XslCompiledTransform
C) A DOM editor
D) A SOAP gateway
E) XmlReader

210. What is the main limitation of XmlReader compared with tree-based APIs?

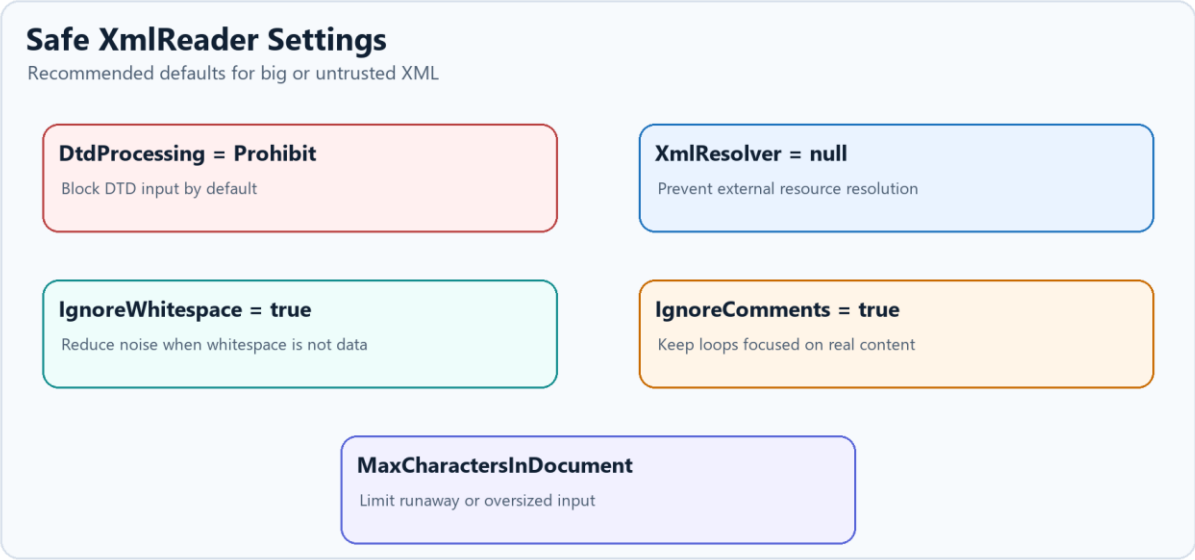
A) It cannot read attributes from the current element
B) It cannot read text nodes from streamed XML content
C) It is forward-only and not suitable for random-access editing
D) It works only with DTD-validated XML documents
E) It requires HTML output before parsing can begin

211. If the task requires random access or editing, what did the lecture recommend?

A) Switch to XmlDocument or XDocument for random access or editing
B) Keep XmlReader and store every node in comments

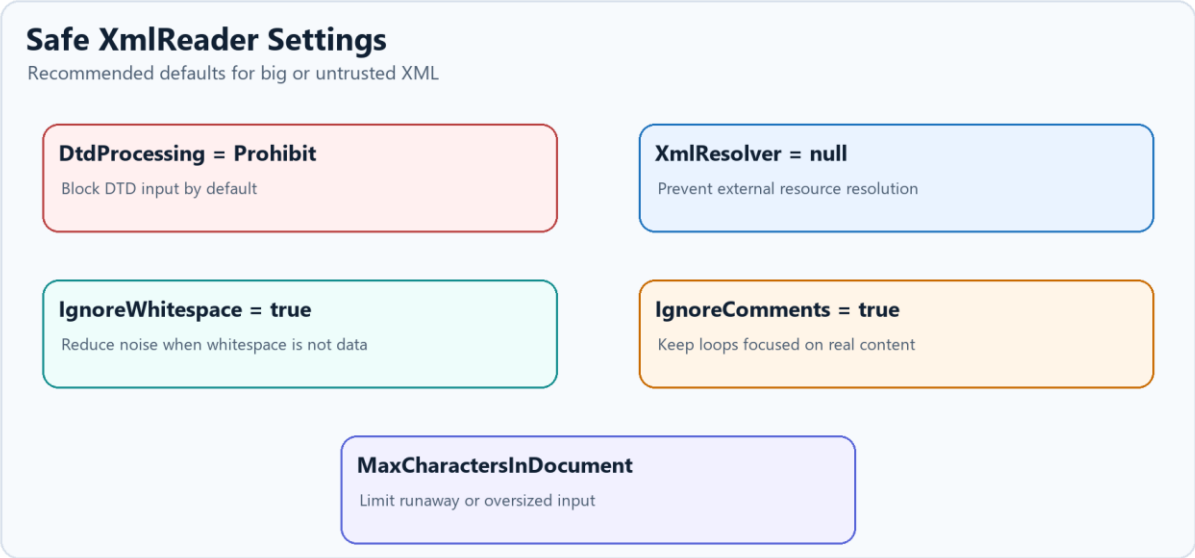
C) Disable namespaces and retry the same streaming logic
D) Convert the XML to CSV immediately before processing
E) Use only DTD validation and ignore tree-based APIs

212. What does DtdProcessing = Prohibit accomplish in a safe XmlReader configuration?



A) It trims whitespace automatically
B) It blocks DTD processing
C) It converts attributes to elements
D) It enables random access
E) It counts records automatically

213. In the safe XmlReader settings diagram, why set XmlResolver = null when reading large or untrusted XML?



A) To make every node uppercase
B) To force mixed content mode
C) To skip all attributes
D) To block external resource resolution
E) To enable schema inference

214. What is the purpose of IgnoreWhitespace = true when whitespace is not meaningful?

A) It loads external files automatically before parsing starts
B) It turns text into attributes during reader traversal
C) It disables NodeType checks inside the main loop
D) It writes a summary report to disk after parsing
E) It removes insignificant whitespace nodes from the loop

215. What is the practical effect of IgnoreComments = true?

A) It prevents XML from having a root element at all
B) It enables DTD processing for comment-heavy documents
C) It keeps comment noise out of the main data loop
D) It converts comments into text nodes before inspection
E) It makes the reader editable like a DOM structure

216. What is the purpose of MaxCharactersInDocument in safe reader settings?

A) It limits runaway or overly large input size
B) It turns large XML into JSON automatically
C) It disables attributes
D) It sorts elements alphabetically
E) It chooses the output encoding

217. Which property tells you whether the cursor is on an element, text node, end element, and so on?

A) Depth
B) NodeType
C) BaseURI
D) Prefix
E) Settings

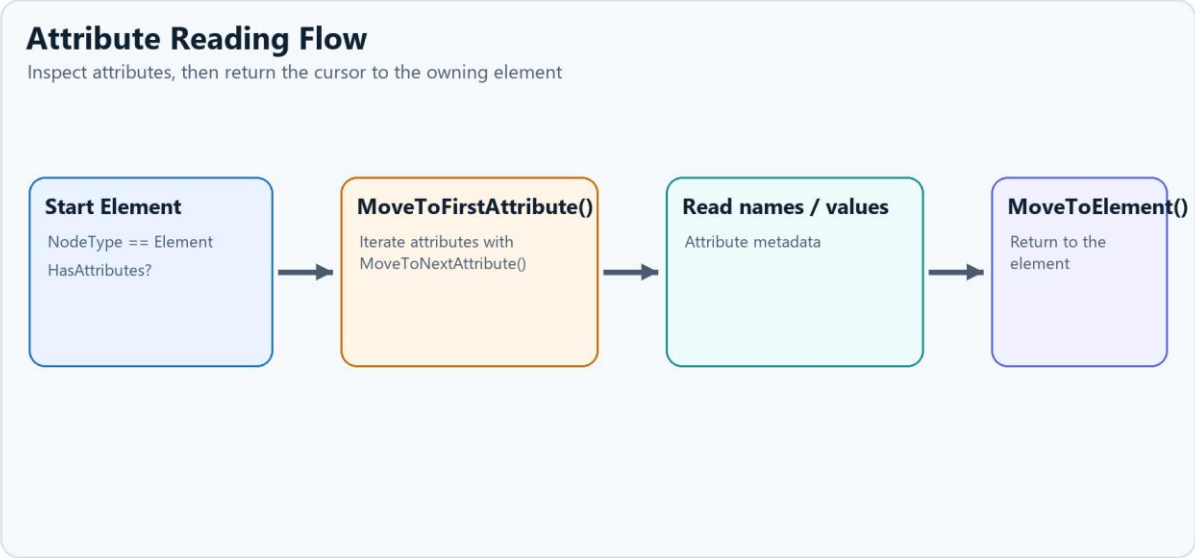
218. What does the Name property provide while streaming?

A) The file size of the XML source
B) The full schema text
C) The current line ending style
D) The current element or attribute name
E) The output report path

219. What does the Depth property help you understand?

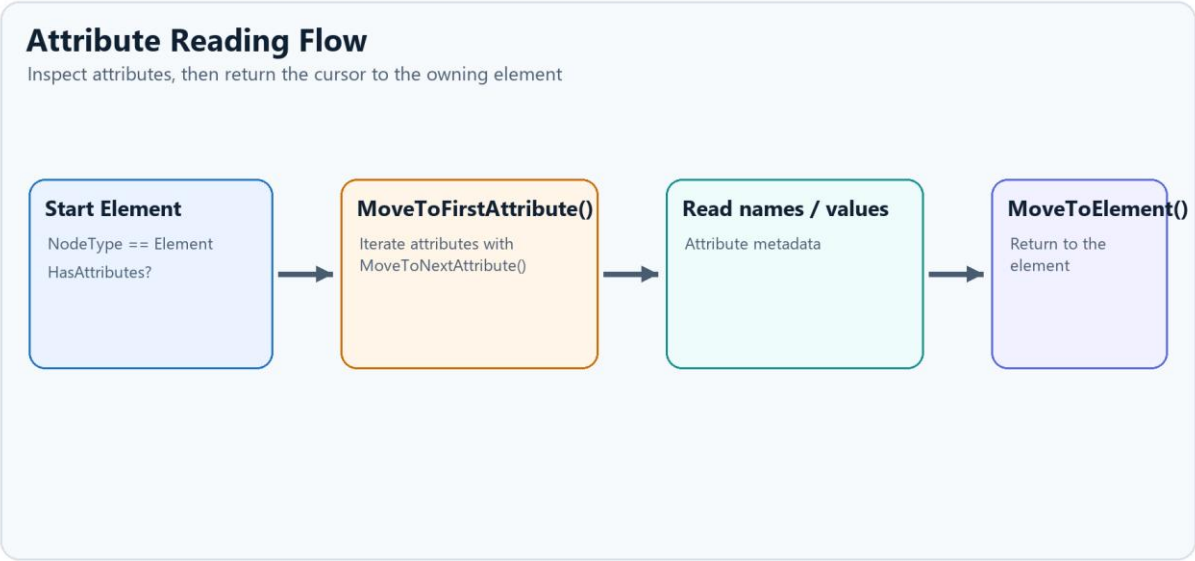
A) The encoding of the writer output
B) The number of schemas loaded
C) The byte size of the current value
D) The HTTP port of the service
E) The nesting level of the current node

220. In the attribute reading flow diagram, which set of members is used to inspect attributes on the current element?



A) ReadOuterXml and WriteNode for full subtree serialization
B) Load and Save for whole-document DOM editing
C) HasAttributes, MoveToFirstAttribute(), and MoveToNextAttribute()
D) XPathSelectElements and Descendants for tree-based navigation
E) Transform and Compile for XSLT processing

221. What important step should follow attribute iteration in a streaming loop?



A) Call MoveToElement() to return to the owning element
B) Reopen the XML file from disk before the next loop step
C) Switch to JSON serialization for the remaining nodes
D) Reset the reader to the document start and parse again
E) Disable namespaces permanently for the current feed

222. Why can ReadElementContentAsString() be dangerous inside a blind while (reader.Read()) loop?

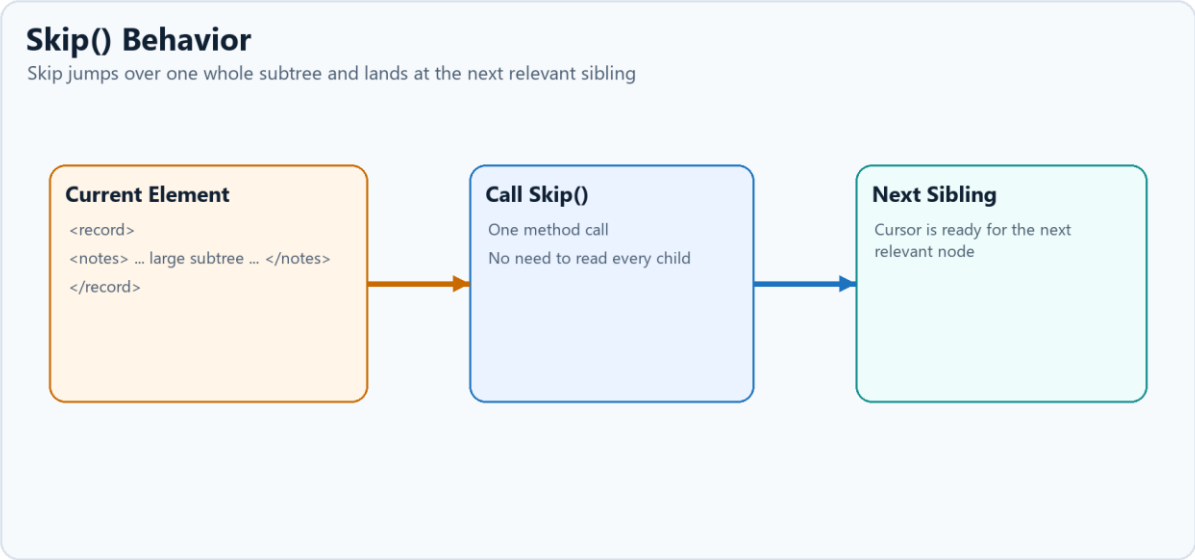
A) It deletes the parent element from the streaming input
B) It advances the reader and can skip siblings
C) It disables NodeType checks for later nodes

D) It requires XSLT to continue the parsing loop
E) It blocks mixed content forever in the XML source

223. For which situation is Skip() especially useful?

A) When every subtree must always be read fully and logged
B) When only the XML declaration is needed from the file
C) When the file must be transformed with XSLT first
D) When irrelevant branches should be bypassed
E) When namespaces need to be declared before parsing

224. What does Skip() do to the reader position?



A) It rewinds to the parent root before the next read
B) It freezes the cursor in place until Read() is called
C) It converts the subtree to a string only for logging
D) It moves into every descendant manually one by one
E) It moves past the current subtree to the node that follows it

225. What is mixed content in XML?

```
<announcement>Welcome <em>XML</em> students.</announcement>
```

A) A document with two root elements at the top level
B) An element that contains only attributes and no children
C) An element that contains both text and child elements
D) An external DTD subset attached to the document
E) A namespace conflict between two element prefixes

226. Why is ReadElementContentAsString() the wrong tool for mixed content?

A) It expects simple text-only content and cannot handle mixed content
B) It can never read plain text elements from any XML source
C) It removes all namespaces from the file before parsing
D) It works only with DTD validation and not plain XML
E) It converts every element to an attribute automatically

227. Which metric was explicitly listed as part of the Week 7 feed processor summary?

A) Average browser refresh time

B) Total course records seen
C) Number of CSS rules used
D) Count of SQL joins executed
E) Number of WSDL operations

228. Which additional feed processor result was highlighted in the lecture?

A) All TCP ports that were opened
B) The total number of DLL files in the project
C) A JSON schema for each record
D) Total available seats across open records
E) The SHA-256 hash of the XML file

229. Which requirement was part of the Week 7 lab design?

A) Build a full web browser from XML
B) Avoid safe settings so students see raw behavior
C) Replace XmlReader with XSLT only
D) Use only screenshots instead of output files
E) Compute at least three summary statistics

230. Which action was explicitly required in the Week 7 lab?

A) Generate a WSDL file
B) Train a classifier on XML tags
C) Write a text report to disk
D) Replace XML with PDF
E) Create a relational database schema

231. Which item was suggested as a strong extension idea rather than a core requirement?

A) Add a second filtered output file with only accepted records
B) Stop using XmlReaderSettings entirely in the feed processor
C) Remove summary statistics from the final report output
D) Delete all attributes before parsing any record data
E) Use only malformed XML as the input data source

232. Which advanced extension idea broadens the parser beyond the simplest feed shape?

A) Converting the report to an image sprite sheet
B) Supporting a namespace-aware feed variant
C) Removing all NodeType checks
D) Using only comments as output
E) Skipping every record unconditionally

233. What was described as the best first debugging move when XmlReader logic behaves strangely?

A) Immediately rewrite the code in SQL and compare outputs
B) Delete the XML file and regenerate it from scratch
C) Disable all safe settings first to expose more issues
D) Print NodeType, Name, and Depth to understand reader movement
E) Assume the XML syntax is always wrong before debugging

234. What happens if you forget MoveToElement() after reading attributes?

A) The reader upgrades to DOM mode and loads the whole tree
B) The XML becomes invalid immediately during parsing
C) The namespaces are erased from the remaining nodes
D) The output file is deleted before the loop finishes

E) The cursor stays on an attribute and later logic can break

235. Why is using while (reader.Read()) together with Skip() risky if the loop is not carefully controlled?

A) Because Skip() disables attributes forever
B) Because Read() cannot be called on XML
C) Because the next sibling may be skipped
D) Because NodeType becomes unavailable
E) Because it always triggers DTD loading

236. Why did the lecture say many XML problems are input-policy bugs, not only parsing bugs?

A) Because untrusted input needs safe limits, not just parsing
B) Because parsers never inspect XML content or reader settings
C) Because NodeType hides malicious input from the application
D) Because XML cannot contain text nodes in real feeds
E) Because comments always crash readers during streaming

237. What did the lecture recommend grading more heavily than flashy formatting in Week 7 labs?

A) Font choice, color palette, and decorative report styling
B) Correct reader movement, safe settings, and clean summary logic
C) How many namespaces are declared in the sample feed
D) How many blank lines appear in the final text report
E) The number of comments written in Program.cs

238. Why is XmlReader well suited to the Week 7 feed processor demo?

A) It builds a full editable tree for later modifications
B) It removes the need for business logic entirely
C) It works only with tiny XML files and simple records
D) It supports one-pass aggregation and filtering while streaming
E) It guarantees perfect data quality without any checks

239. Which statement best captures when streaming is the wrong tool?

A) When the XML contains attributes on most elements
B) When the XML contains namespaces and qualified names
C) When the file is read from disk instead of a string
D) When the XML has only one root element and many children
E) When the task needs random access or editing

240. Which statement best summarizes the Week 7 lecture?

A) It focused on XSLT grouping and sorting only this week
B) It focused on DTD content models only in validation labs
C) It focused on safe XmlReader streaming and one-pass aggregation
D) It focused on WSDL and SOAP bindings for service work
E) It focused on replacing XML with JSON across the course

Answer Key

1. C 2. A 3. E 4. B 5. D 6. C 7. A 8. E 9. B 10. D 11. C 12. A

13. E 14. B 15. D 16. C 17. A 18. E 19. B 20. D 21. C 22. A 23. E 24. B

25. D 26. C 27. A 28. E 29. B 30. D 31. C 32. A 33. E 34. B 35. D 36. B

37. D 38. A 39. C 40. E 41. B 42. D 43. A 44. C 45. E 46. B 47. D 48. A

49. C 50. E 51. B 52. D 53. A 54. C 55. E 56. B 57. D 58. A 59. C 60. E

61. B 62. D 63. A 64. C 65. E 66. B 67. D 68. A 69. C 70. E 71. A 72. C

73. B 74. D 75. E 76. A 77. C 78. B 79. D 80. E 81. A 82. C 83. B 84. D

85. E 86. A 87. C 88. B 89. D 90. E 91. A 92. C 93. B 94. D 95. E 96. A

97. C 98. B 99. D 100. E 101. A 102. C 103. B 104. D 105. E 106. B 107. A 108. C

109. D 110. E 111. B 112. A 113. C 114. D 115. E 116. B 117. A 118. C 119. D 120. E

121. B 122. A 123. C 124. D 125. E 126. B 127. A 128. C 129. D 130. E 131. B 132. A

133. C 134. E 135. B 136. A 137. C 138. D 139. D 140. A 141. E 142. B 143. C 144. D

145. A 146. E 147. B 148. C 149. D 150. A 151. E 152. B 153. C 154. D 155. A 156. E

157. B 158. C 159. D 160. A 161. E 162. B 163. C 164. D 165. A 166. E 167. B 168. D

169. A 170. E 171. B 172. C 173. C 174. E 175. A 176. D 177. B 178. C 179. E 180. A

181. D 182. B 183. C 184. E 185. A 186. D 187. B 188. C 189. E 190. A 191. D 192. B

193. C 194. E 195. A 196. D 197. B 198. C 199. E 200. A 201. D 202. C 203. E 204. A

205. D 206. B 207. B 208. D 209. E 210. C 211. A 212. B 213. D 214. E 215. C 216. A

217. B 218. D 219. E 220. C 221. A 222. B 223. D 224. E 225. C 226. A 227. B 228. D

229. E 230. C 231. A 232. B 233. D 234. E 235. C 236. A 237. B 238. D 239. E 240. C